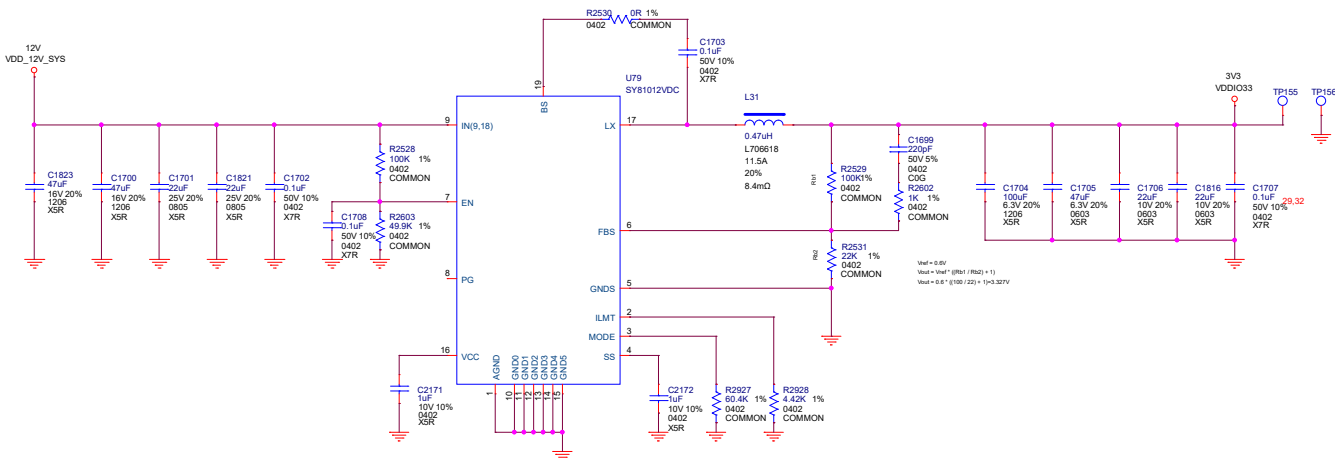
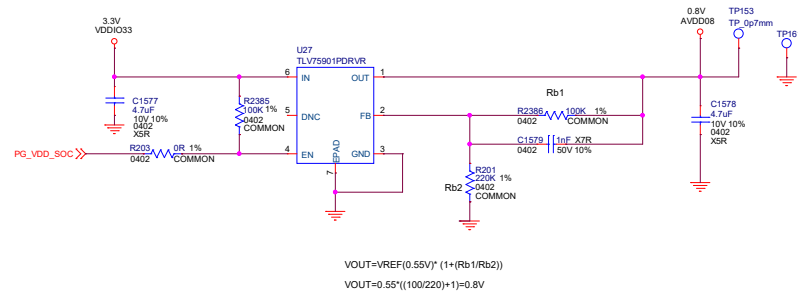


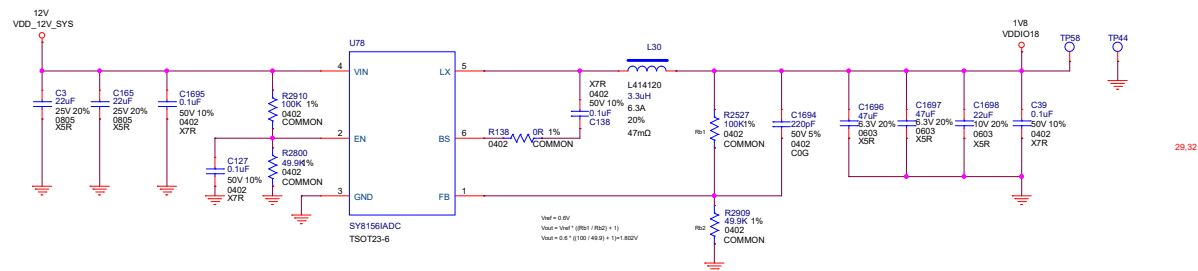
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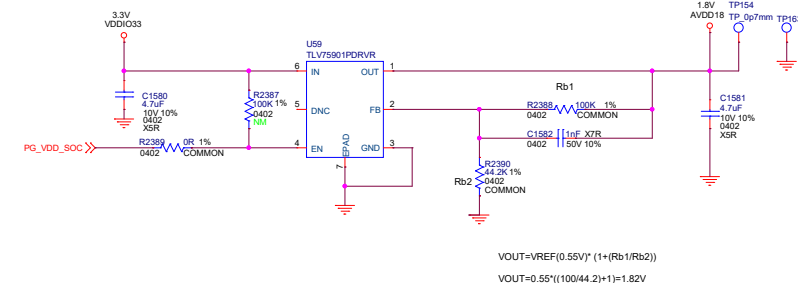
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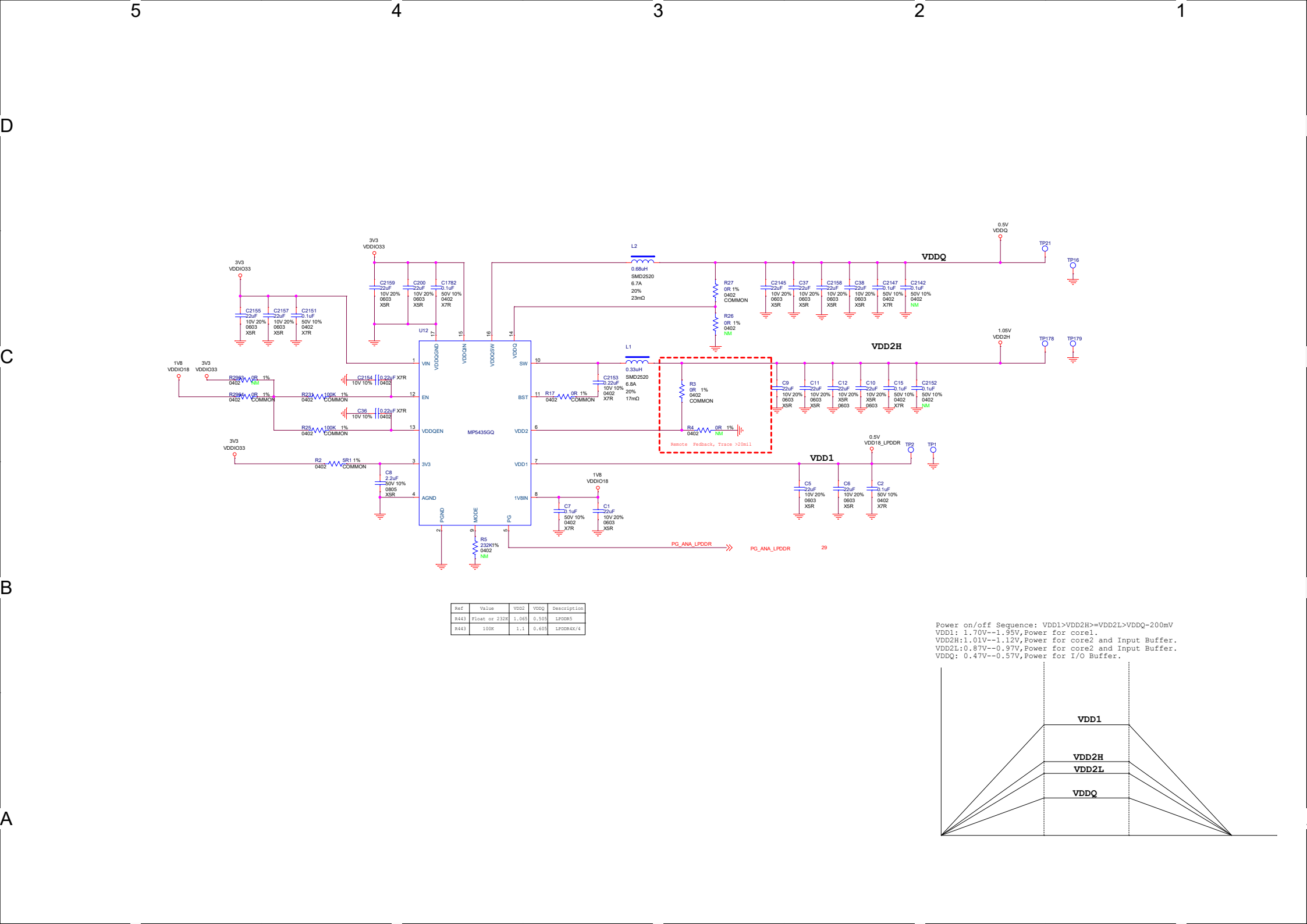


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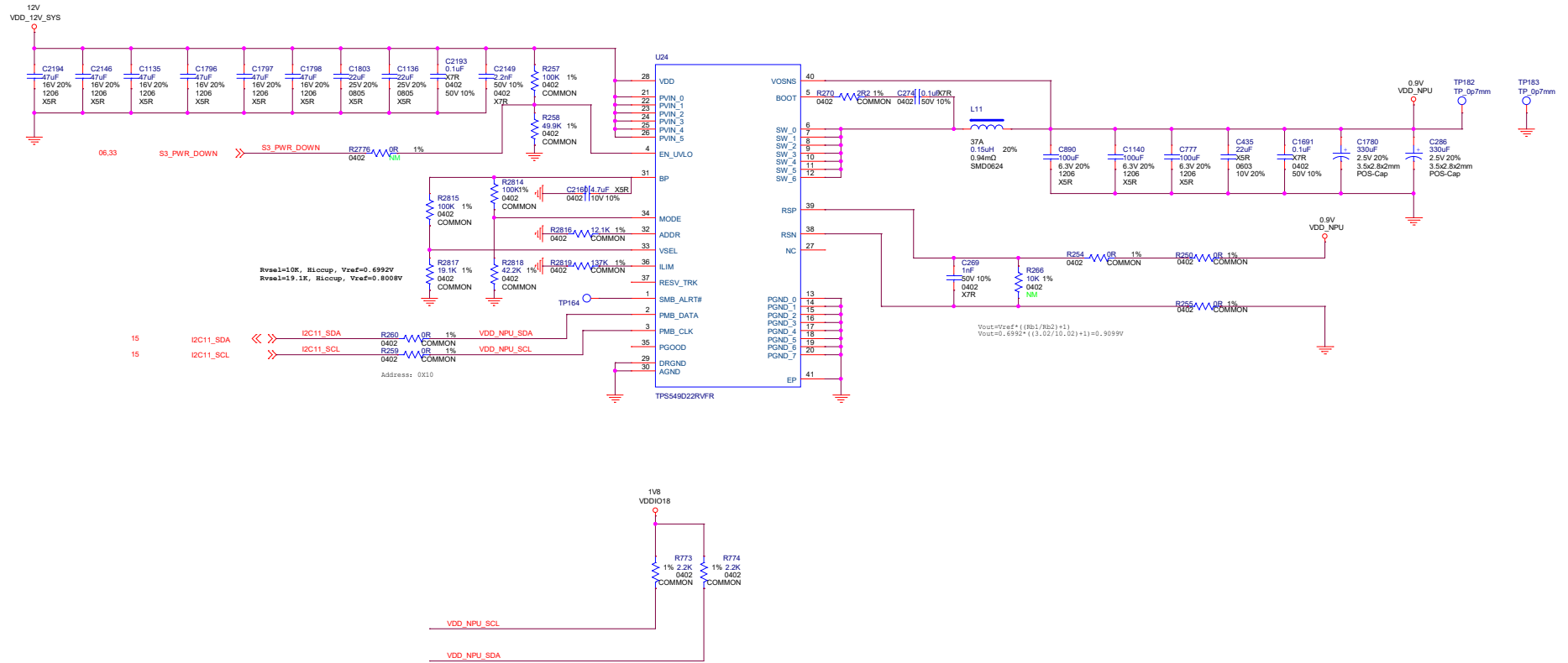


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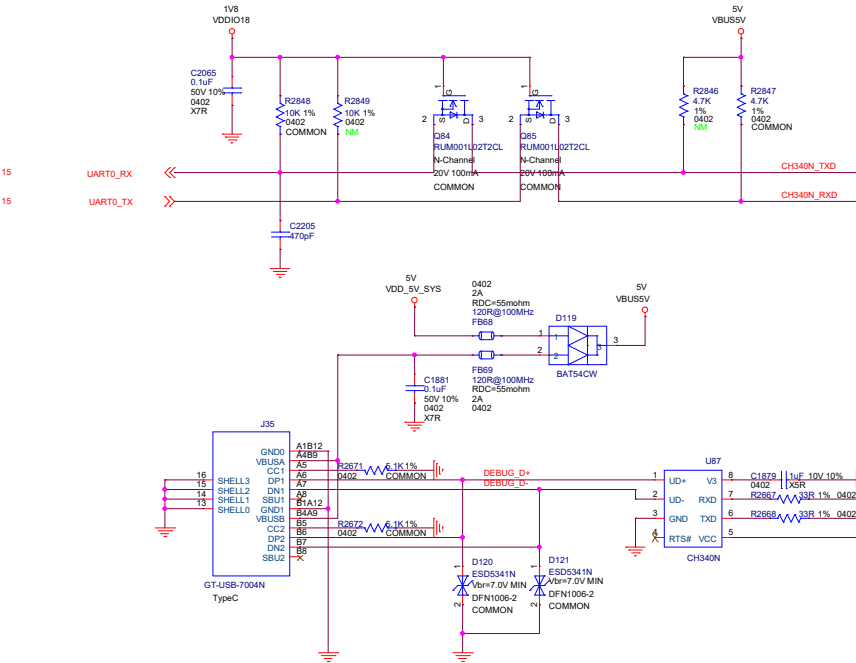




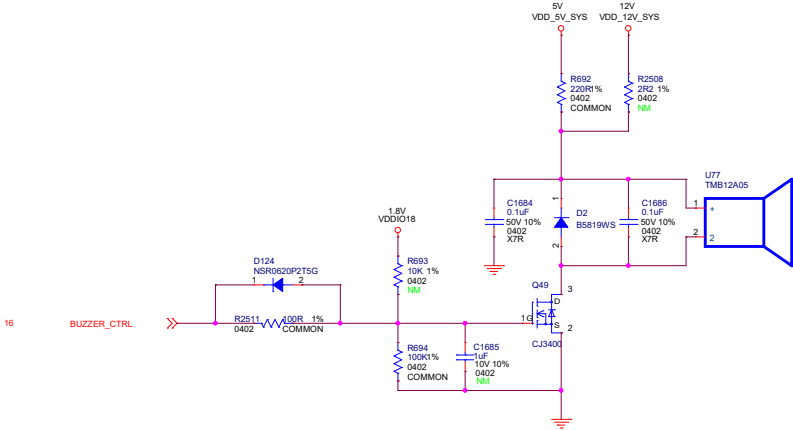
A



Debug UART



Buzzer



D

C

B

A

5

GPIO GOURP A 32BIT WITH INTERRUPT

GPIO Number	FUNCTION IN DESIGN	DIRECTION	ACTIVE
GPIO00	POWER POWER OFF	Input/Output	Low : Power Off High: Normal Run
GPIO01	GPIO Function Unused		
GPIO02	GPIO Function Unused		
GPIO03	RTC Interrupt(Reserve)	Input	Active Low
GPIO04	PMU Interrupt Input	Input	Active Low
GPIO05	U.2 Port 0 PCIe Interface Detect	Input/Output	Active Low
GPIO06	USB CC ID	Input	Low : Attach as source High: Attach as sink
GPIO07	GPIO Function Unused		
GPIO08	GPIO Function Unused		
GPIO09	GPIO Function Unused		
GPIO10	GPIO Function Unused		
GPIO11	U.2 Port 1 PCIe Interface Detect	Input/Output	Active Low
GPIO12	USB CS Interrupt	Input	Active Low
GPIO13	SATA Port 1 Interface Detect	Input	Active Low
GPIO14	SATA Port 2 Interface Detect	Input	Active Low
GPIO15	SATA Port 3 Interface Detect	Input	Active Low
GPIO16	SATA Port 4 Interface Detect	Input	Active Low
GPIO17	GPIO Function Unused		
GPIO18	Burn In Test Key	Input	Active Low
GPIO19	Board Test Key	Input	Active Low
GPIO20	Manufacture Test Key	Input	Active Low
GPIO21	GPIO Function Unused		
GPIO22	GPIO Function Unused		
GPIO23	SOC Fan Tach(Reserve)	Input	Active Low
GPIO24	GPIO Function Unused		
GPIO25	Temperature Interrupt(Reserve)	Input	Active Low
GPIO26	Factory Default Key	Input	Active Low
GPIO27	GPIO Function Unused		
GPIO28	SOC Voltage Adjust	Output	HIGH: 0.9V LOW: 0.814V
GPIO29	GPIO Function Unused		
GPIO30	GPIO Function Unused		
GPIO31	GPIO Function Unused		

4

GPIO GROUP B 32BIT

GPIO Number	FUNCTION IN DESIGN	DIRECTION	ACTIVE
GPIO32	GPIO Function Unused		
GPIO33	GPIO Function Unused		
GPIO34	GPIO Function Unused		
GPIO35	Power LED Select	Output	HIGH: WHITE LED LOW: RED LED
GPIO36	GPIO Function Unused		
GPIO37	GPIO Function Unused		
GPIO38	GPIO Function Unused		
GPIO39	GPIO Function Unused		
GPIO40	GPIO Function Unused		
GPIO41	Power LED CTRL	Output	HIGH:LED ON LOW: LED OFF
GPIO42	GPIO Function Unused		
GPIO43	GPIO Function Unused		
GPIO44	GPIO Function Unused		
GPIO45	GPIO Function Unused		
GPIO46	GPIO Function Unused		
GPIO47	GPIO Function Unused		
GPIO48	GPIO Function Unused		
GPIO49	GPIO Function Unused		
GPIO50	GPIO Function Unused		
GPIO51	GPIO Function Unused		
GPIO52	GPIO Function Unused		
GPIO53	GPIO Function Unused		
GPIO54	GPIO Function Unused		
GPIO55	GPIO Function Unused		
GPIO56	GPIO Function Unused		
GPIO57	GPIO Function Unused		
GPIO58	GPIO Function Unused		
GPIO59	GPIO Function Unused		
GPIO60	GPIO Function Unused		
GPIO61	GPIO Function Unused		
GPIO62	GPIO Function Unused		
GPIO63	GPIO Function Unused		

3

GPIO GROUP C 32BIT

GPIO Number	FUNCTION IN DESIGN	DIRECTION	ACTIVE
GPIO64	GPIO Function Unused		
GPIO65	GPIO Function Unused		
GPIO66	GPIO Function Unused		
GPIO67	GPIO Function Unused		
GPIO68	GPIO Function Unused		
GPIO69	GPIO Function Unused		
GPIO70	MD01 LED Select	Output	HIGH: WHITE LED LOW: RED LED
GPIO71	MD01 LED CTRL	Output	HIGH:LED ON LOW: LED OFF
GPIO72	GPIO Function Unused		
GPIO73	MD02 LED Select	Output	HIGH: WHITE LED LOW: RED LED
GPIO74	MD02 LED CTRL	Output	HIGH:LED ON LOW: LED OFF
GPIO75	GPIO Function Unused		
GPIO76	MD03 LED Select	Output	HIGH: WHITE LED LOW: RED LED
GPIO77	MD03 LED CTRL	Output	HIGH:LED ON LOW: LED OFF
GPIO78	GPIO Function Unused		
GPIO79	MD04 LED Select	Output	HIGH: WHITE LED LOW: RED LED
GPIO80	MD04 LED CTRL	Output	HIGH:LED ON LOW: LED OFF
GPIO81	GPIO Function Unused		
GPIO82	GPIO Function Unused		
GPIO83	GPIO Function Unused		
GPIO84	GPIO Function Unused		
GPIO85	GPIO Function Unused		
GPIO86	GPIO Function Unused		
GPIO87	GPIO Function Unused		
GPIO88	GPIO Function Unused		
GPIO89	GPIO Function Unused		
GPIO90	GPIO Function Unused		
GPIO91	GPIO Function Unused		
GPIO92	GPIO Function Unused		
GPIO93	GPIO Function Unused		
GPIO94	PMU Reset	Output	Active Low
GPIO95	GPIO Function Unused		

2

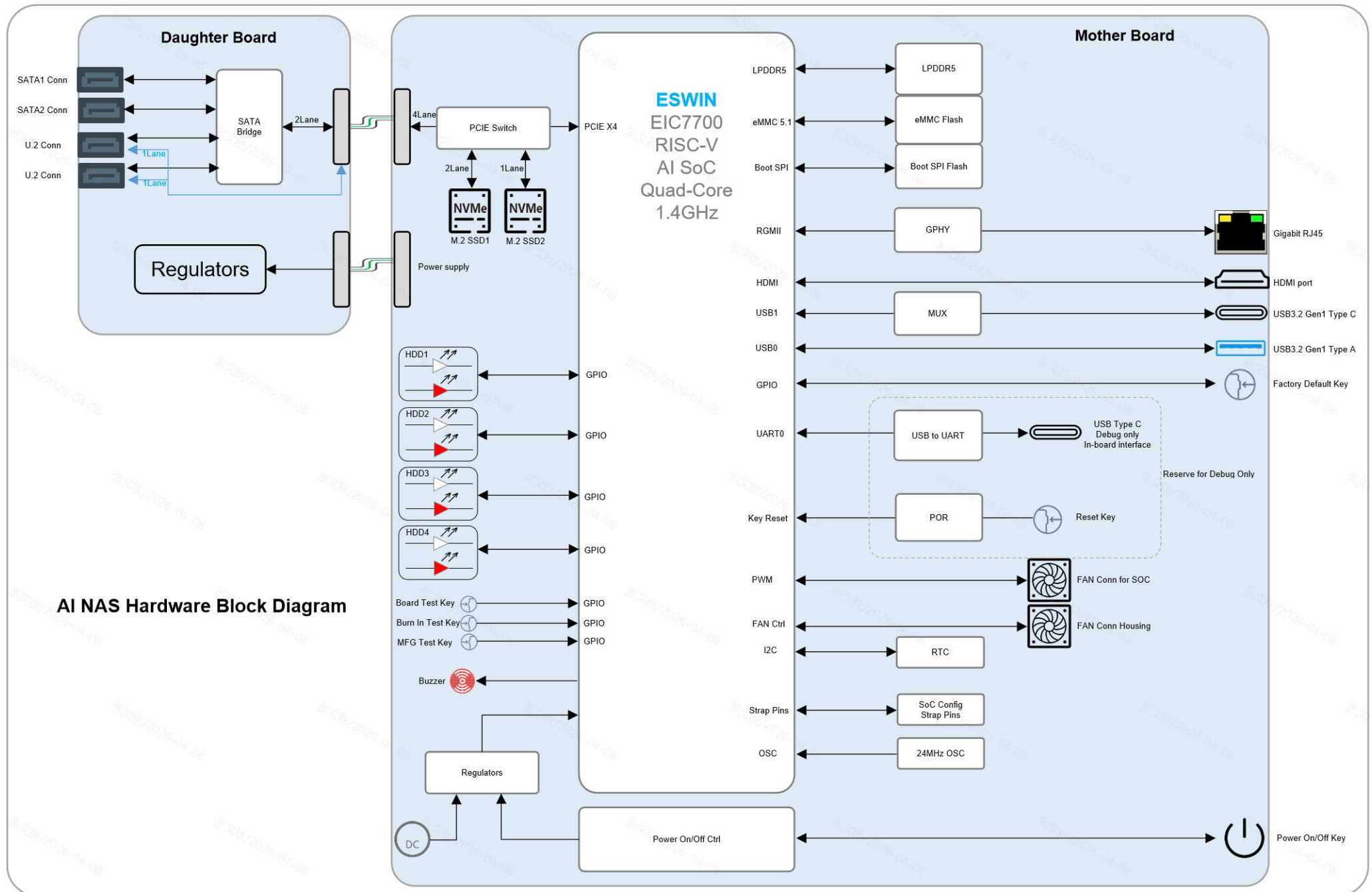
GPIO GROUP D 16BIT

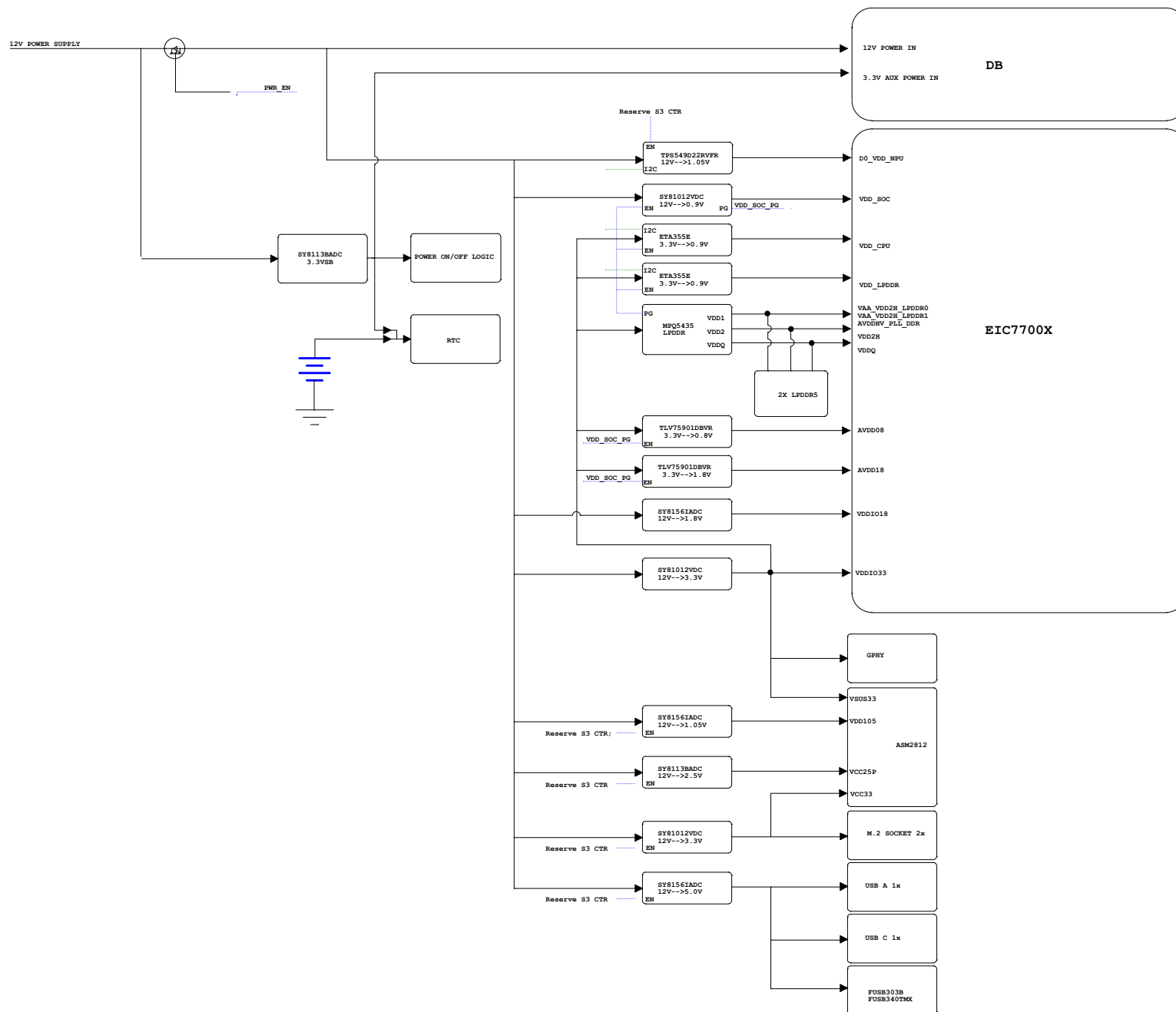
GPIO Number	FUNCTION IN DESIGN	DIRECTION	ACTIVE
GPIO96	GPIO Function Unused		
GPIO97	GPIO Function Unused		
GPIO98	GPIO Function Unused		
GPIO99	GPIO Function Unused		
GPIO100	GPIO Function Unused		
GPIO101	GPIO Function Unused		
GPIO102	GPIO Function Unused		
GPIO103	GPIO Function Unused		
GPIO104	GPIO Function Unused		
GPIO105	GPIO Function Unused		
GPIO106	Buzzer CTRL	Output	Active High
GPIO107	GPIO Function Unused		
GPIO108	GPIO Function Unused		
GPIO109	GPIO Function Unused		
GPIO110	GPIO Function Unused		
GPIO111	S3 Power Save Mode(Reserve)	Output	HIGH: Power Save Mode LOW: Normal Run

1

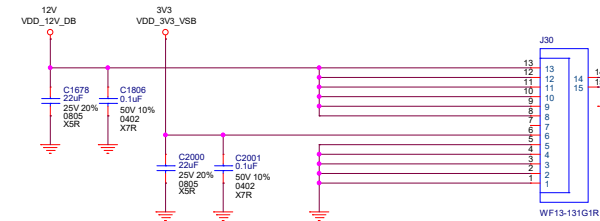
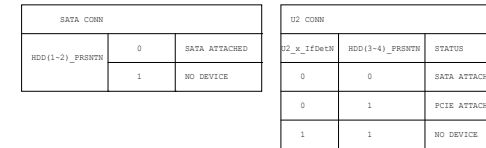
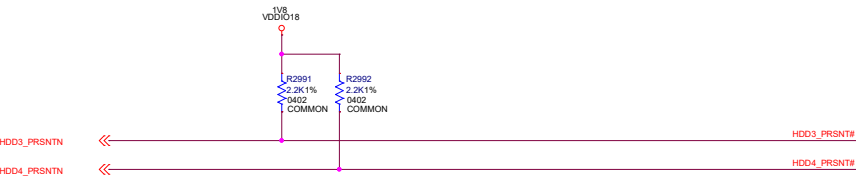
SOC I2C MAP

Port	Slave Device	Slave Addr	COMMENTS
I2C0	ETA355E	0xC0	VDD_CPU PMIC
I2C1	ETA355E	0xC0	VDD_LPDDR PMIC
I2C2	PCF85063ATL/1,118	7'b1010001	RTC
I2C3	NOT USED		
I2C4	NOT USED		
I2C5	TMP102AIDRLR	7'b100_1000	Temperature Sensor
I2C6	NOT USED		
I2C7	NOT USED		
I2C8	NVME SSD2	NA	Address depends on m.2 card
I2C9	NVME SSD1	NA	Address depends on m.2 card
I2C10	NOT USED		
I2C11	TPS549D22RVFR	7'b001_0000	VDD_NPU PMIC

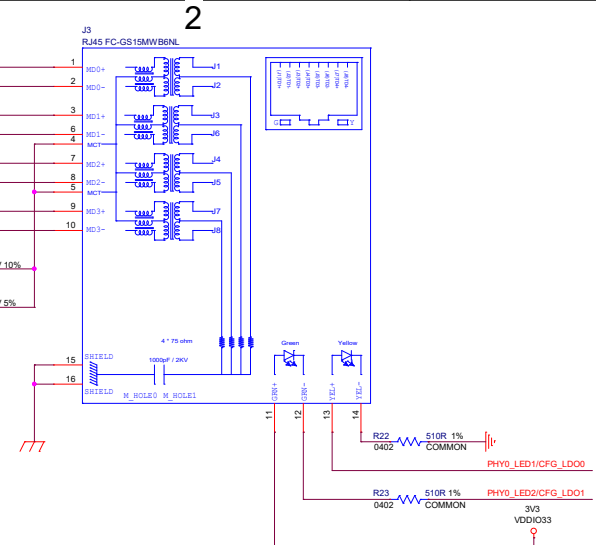
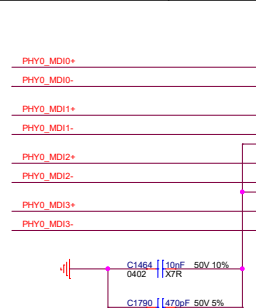
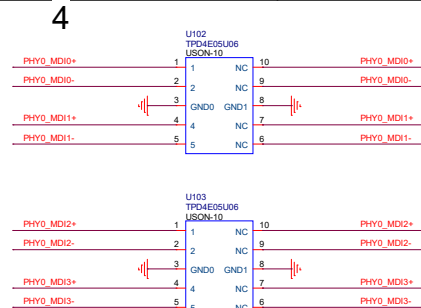
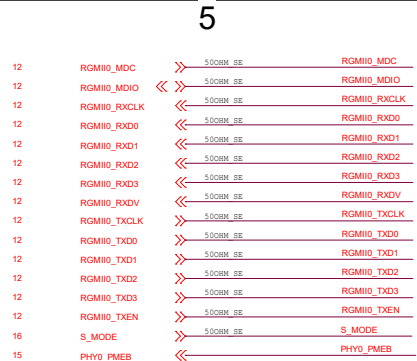




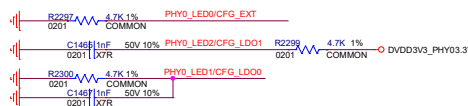
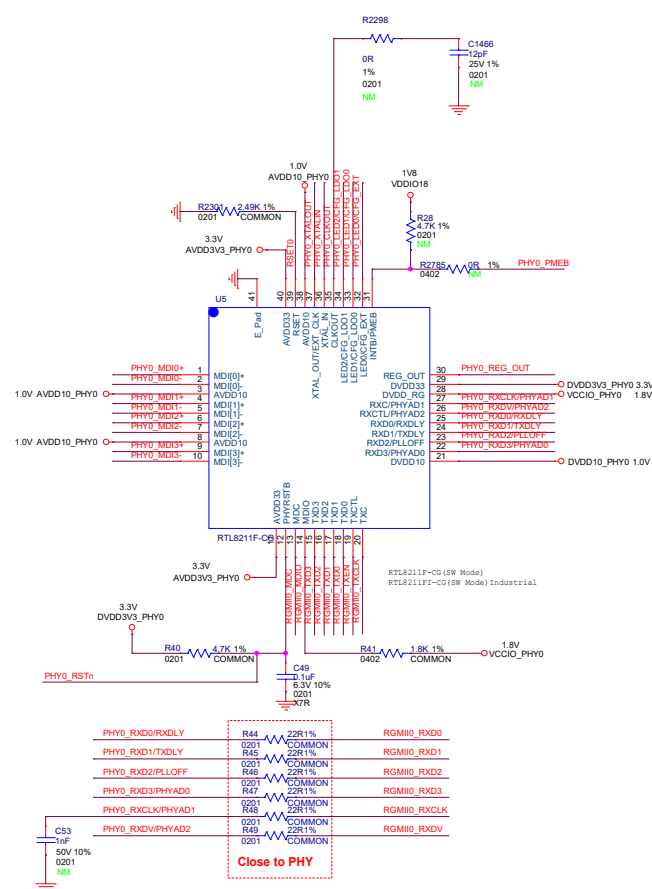
A



D



C



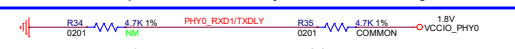
VCC_PHY0_IO Voltage Config			
RGMII Power Source	CFG_EXT	CFG_LDO1(1:0)	COMMENTS
External 3.3V	1'b1	2'b00	CFG_EXT: 1: External Power Source for IO pad. 0: Integrated LDO for IO pad.
External 1.8V	1'b1	2'b10	CFG_LDO1(1:0): 1: 1.8V 2: 3.3V
Internal 1.8V (default)	1'b0	2'b10	



PHY Address Config 000



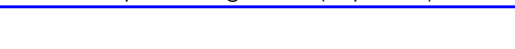
Pull-up for additional 2ns delay to RXC for data latching



Pull-up for additional 2ns delay to TXC for data latching



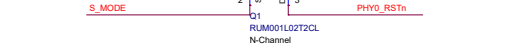
Pull-up to disable PLL @ ALDPS mode(Low power mode)



S_MODE



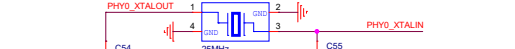
PHY0_RSTn



PHY0_RSTn



PHY0_RSTn

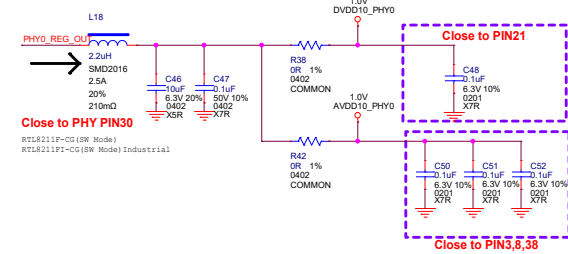


PHY0_RSTn



PHY0_RSTn

B



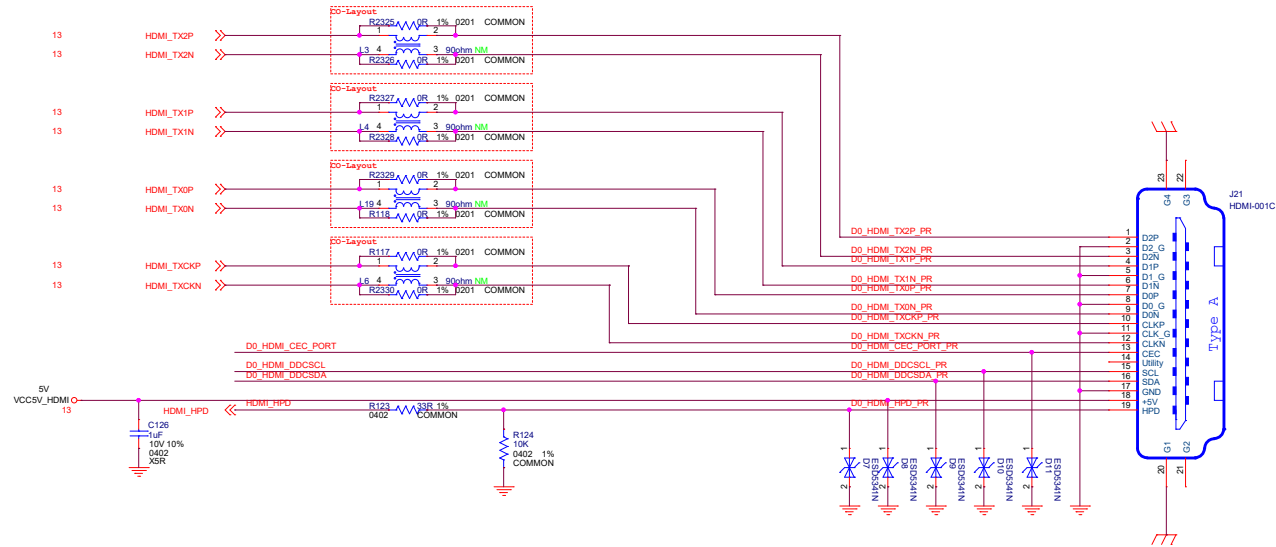
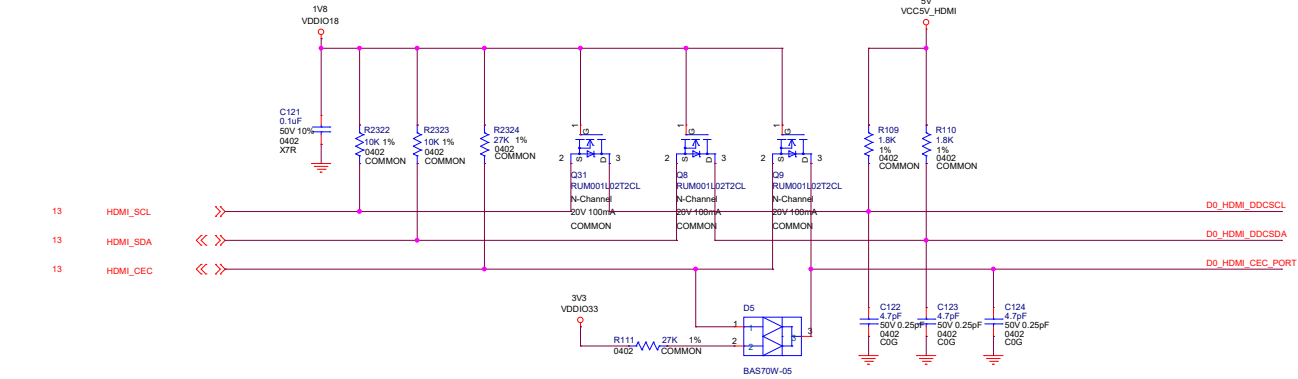
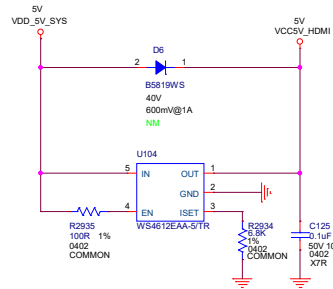
A

D

C

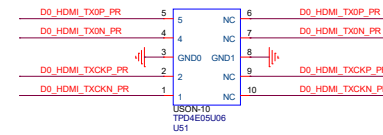
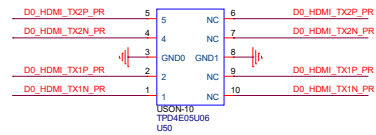
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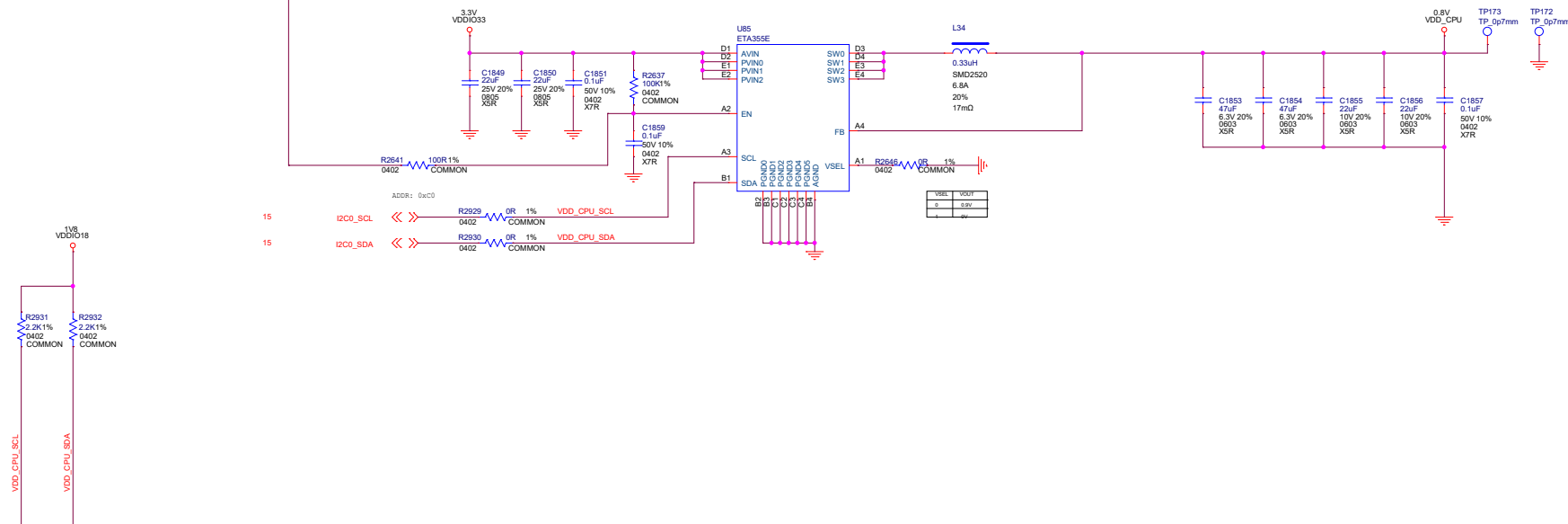
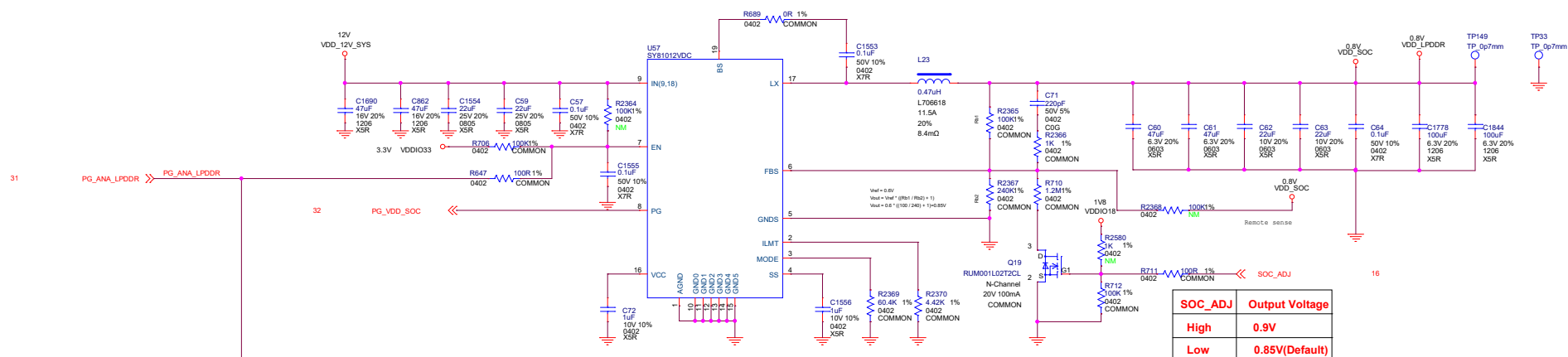
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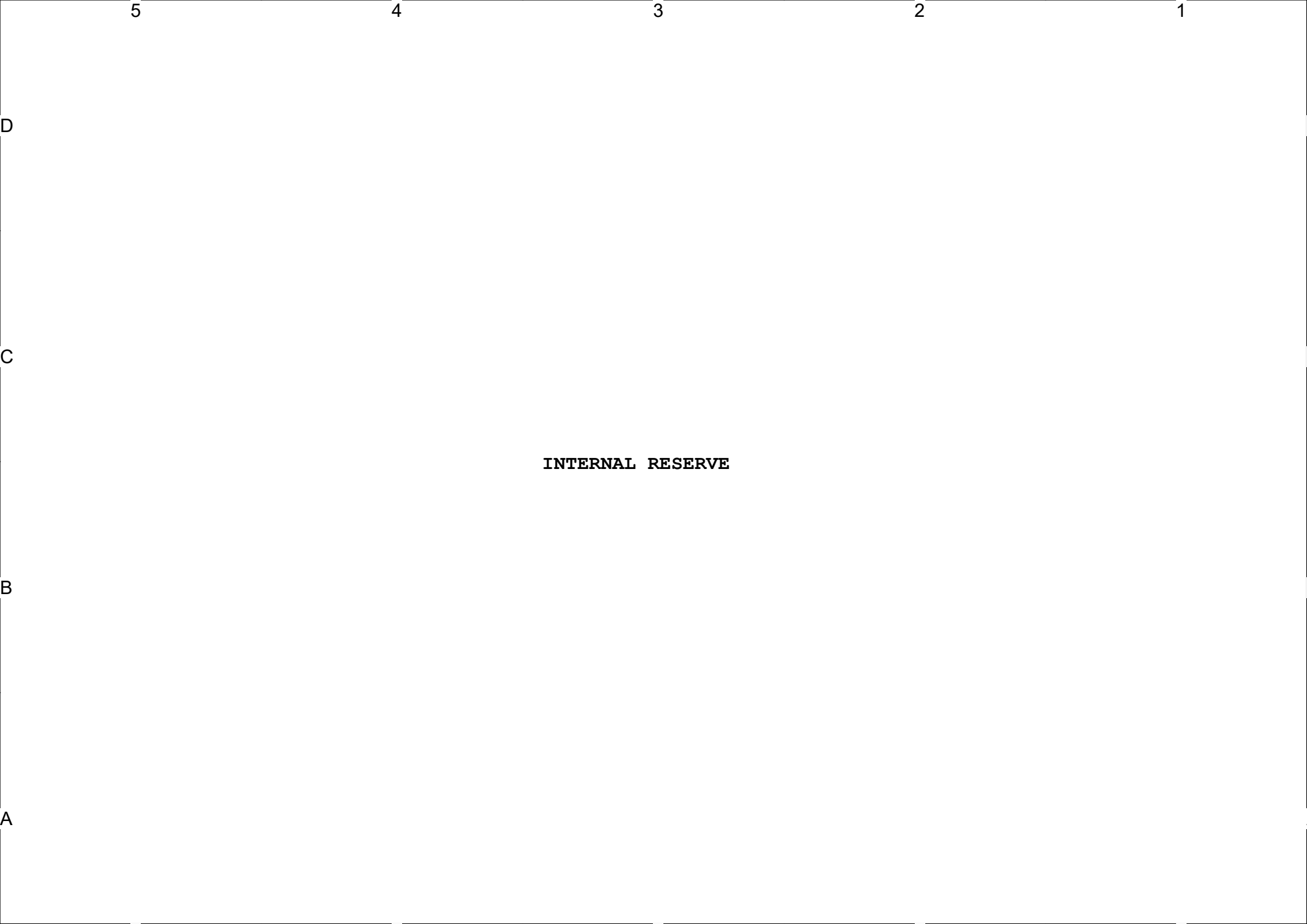


ESD

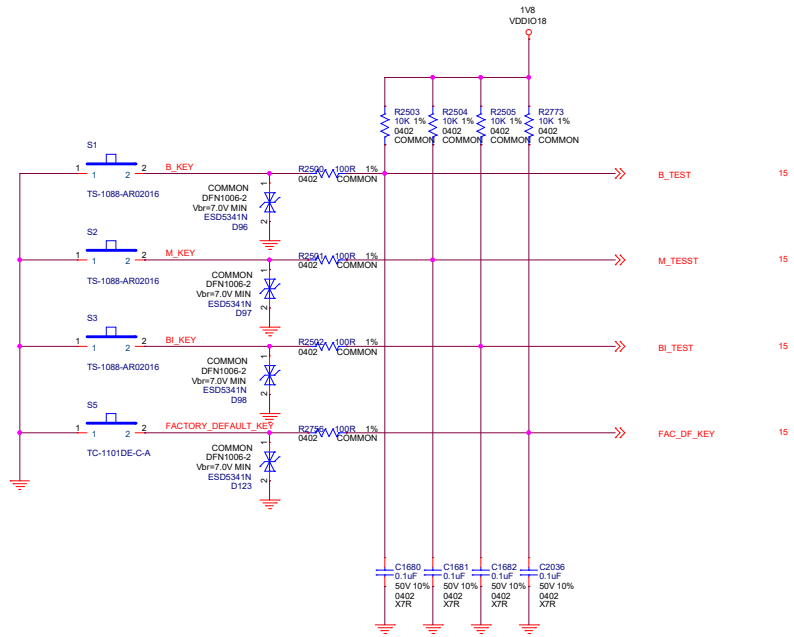
Note: All the ESD components should be placed close to the port and $C_j < 0.4\text{pF}$



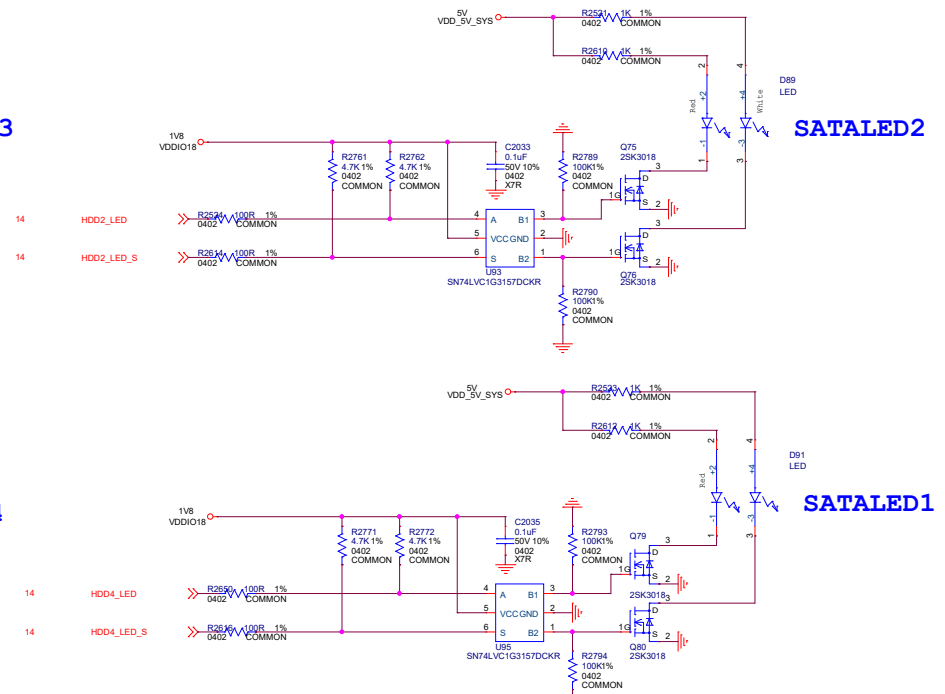
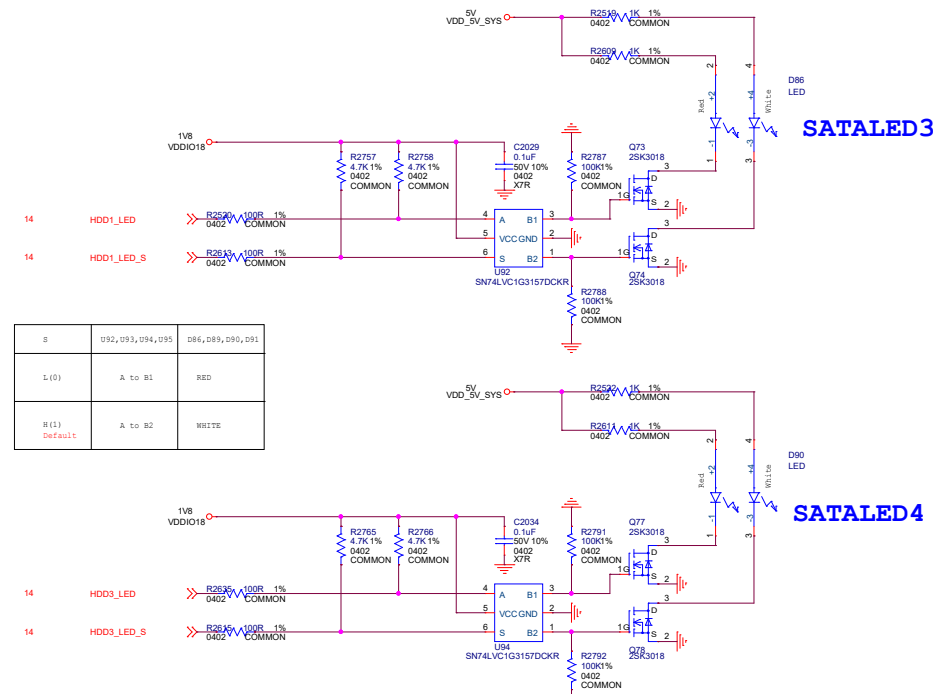
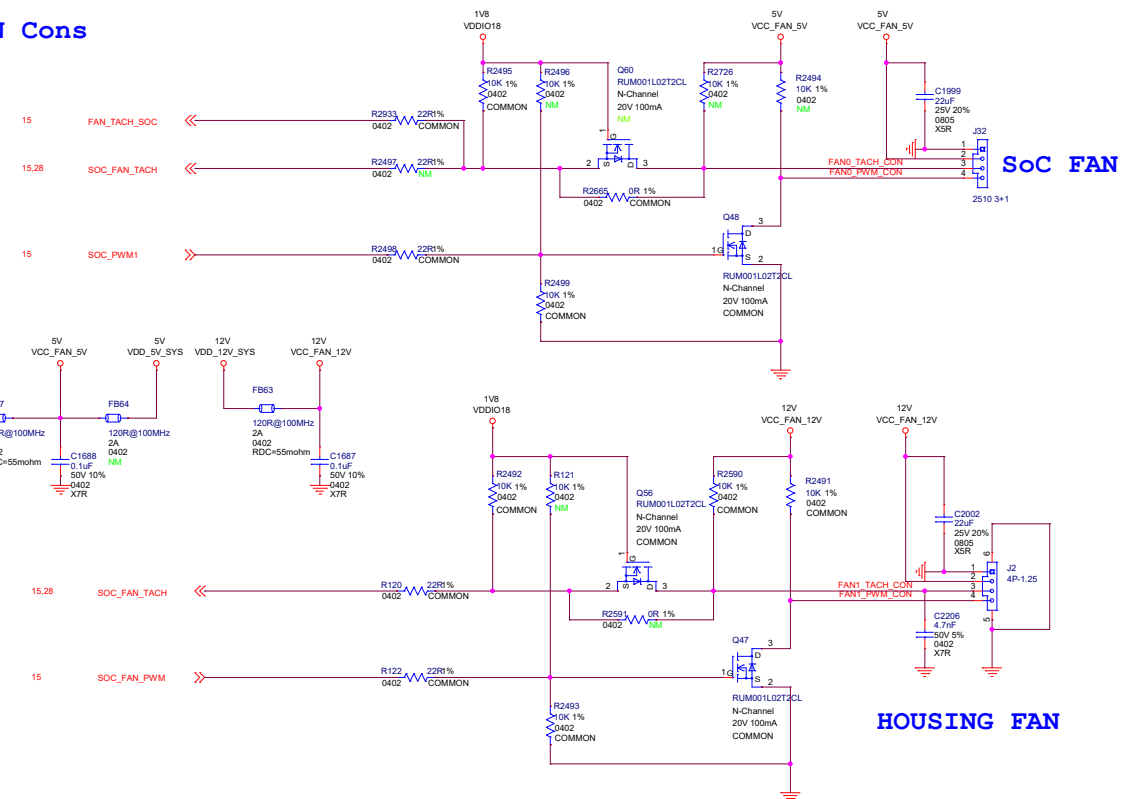


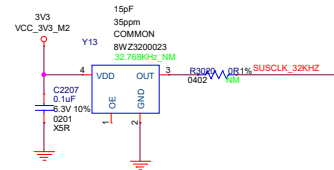
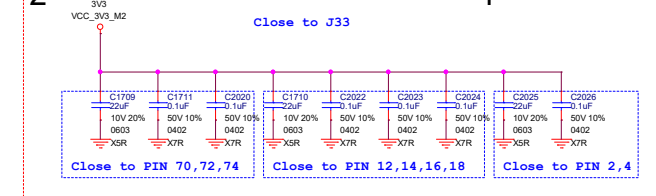


KEYS

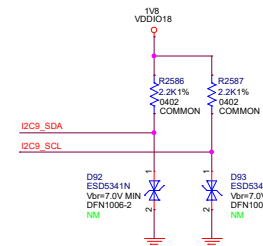
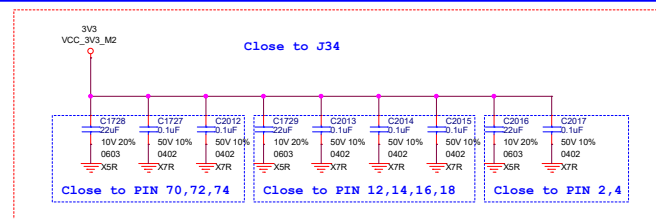


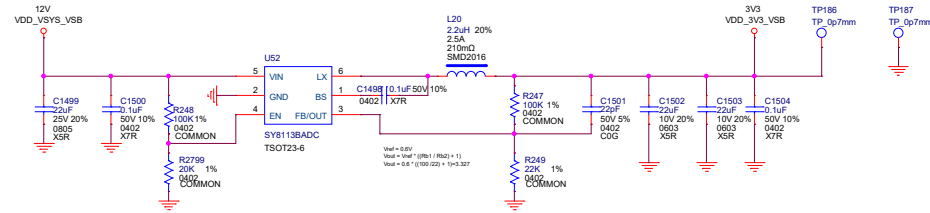
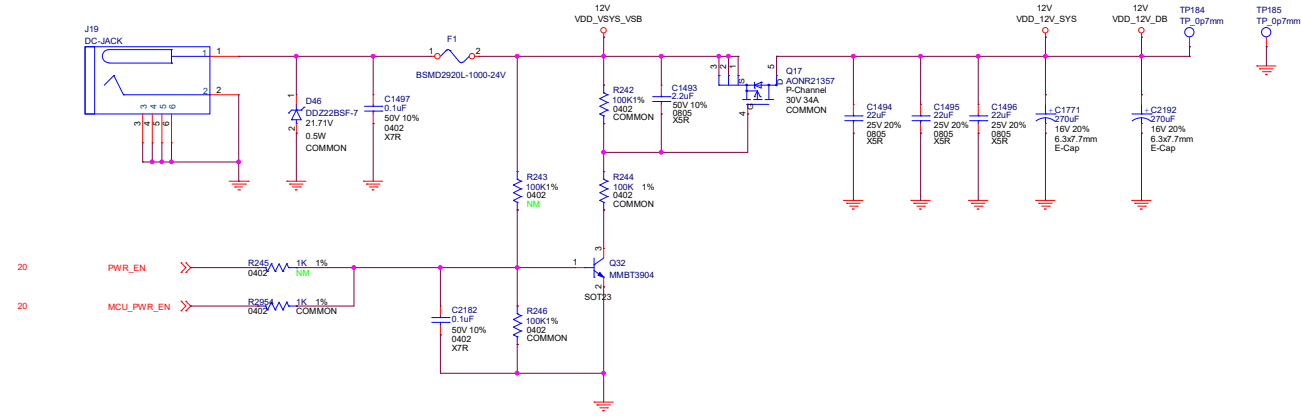
FAN Cons



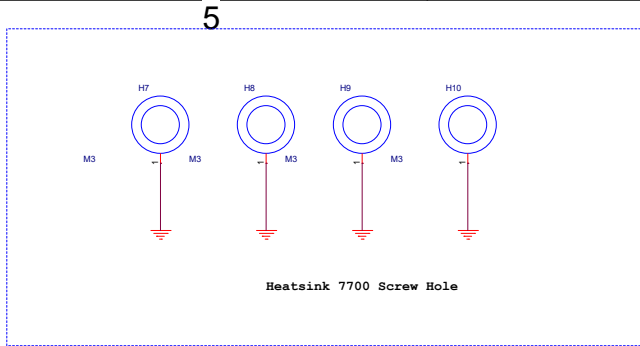


SS1	SS0	Spread(%)
0	0	No Spread
0	1	-0.5
1	0	-0.75
1	1	No Spread

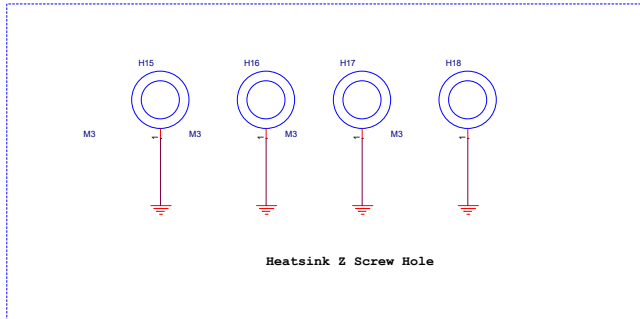




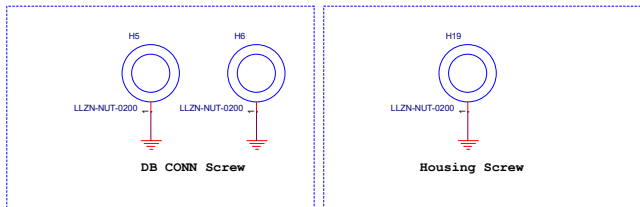
D



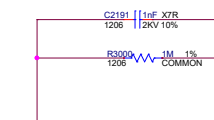
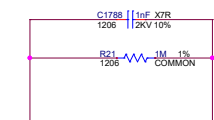
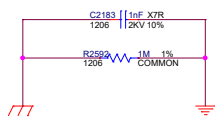
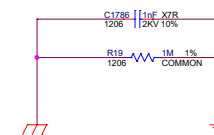
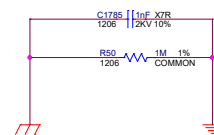
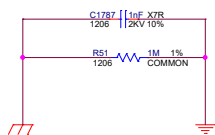
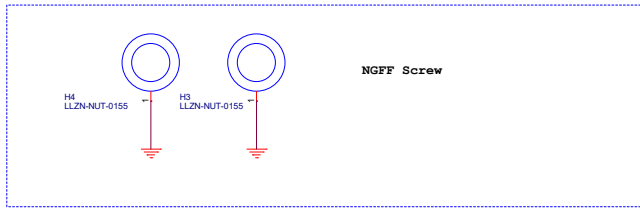
C



B



A



2

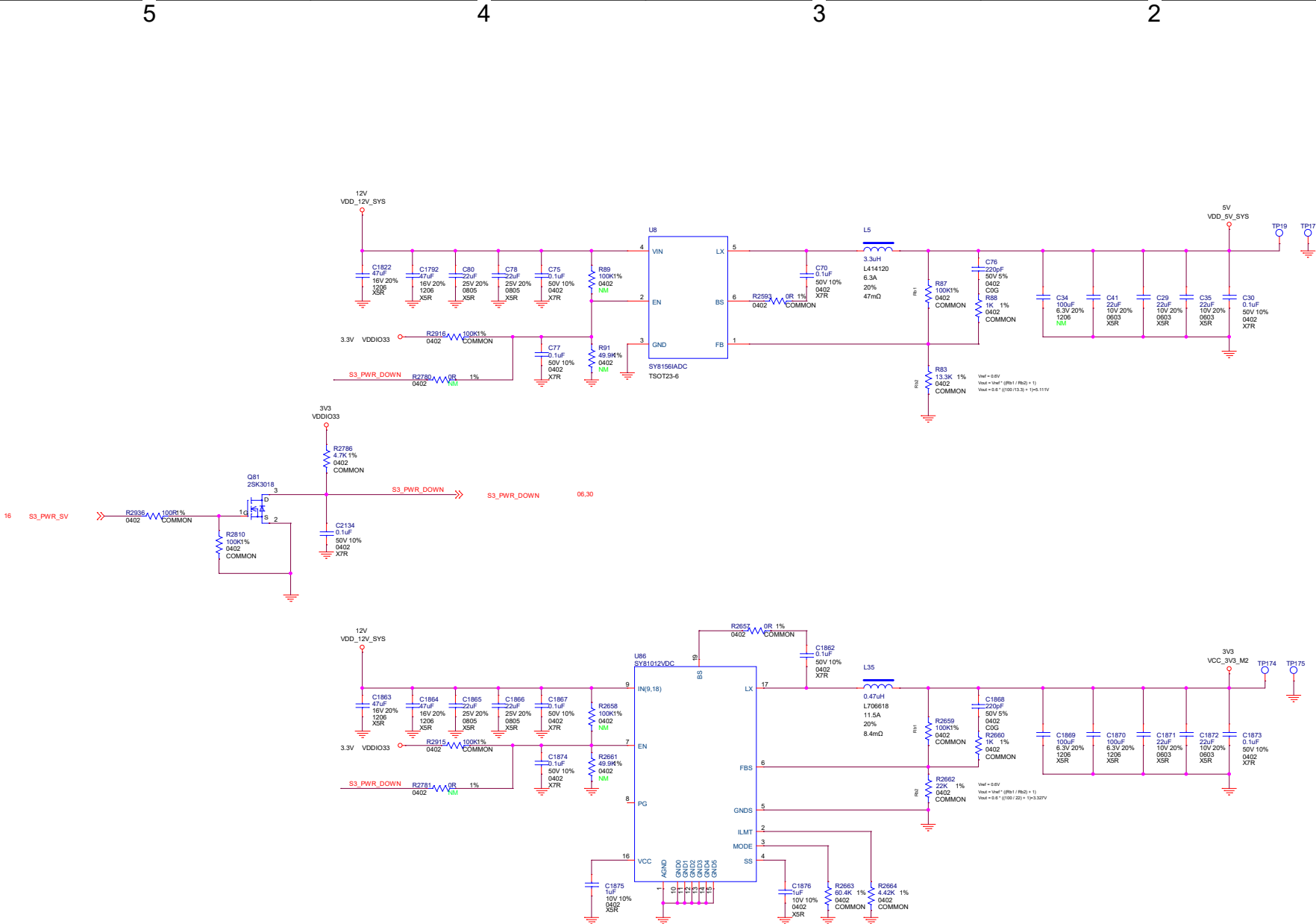
1

D

C

B

A

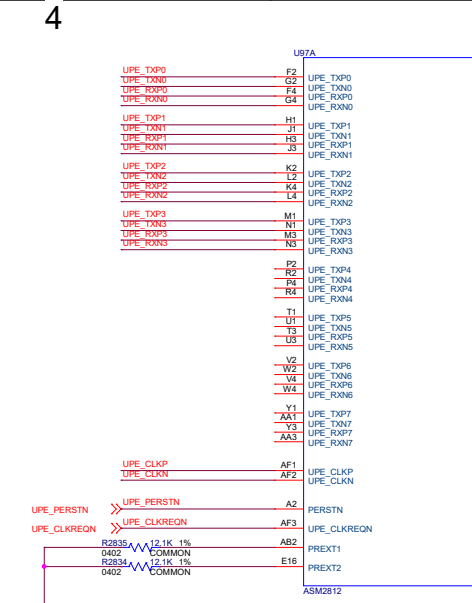
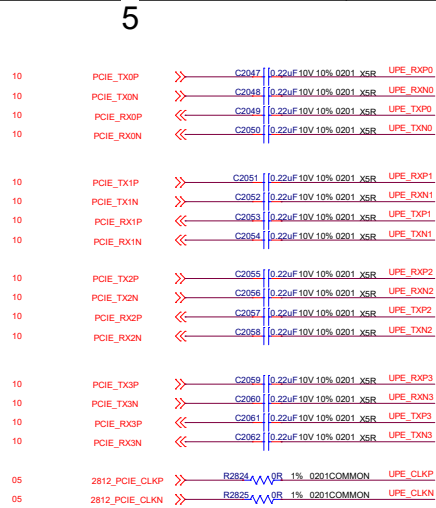


D

C

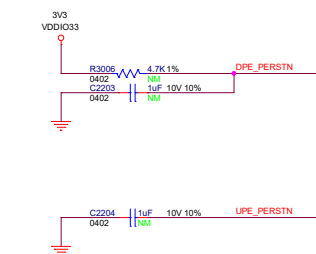
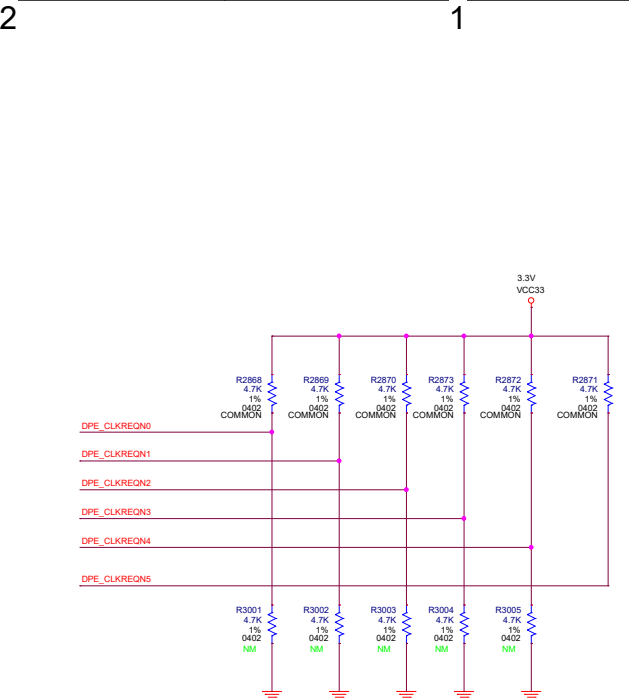
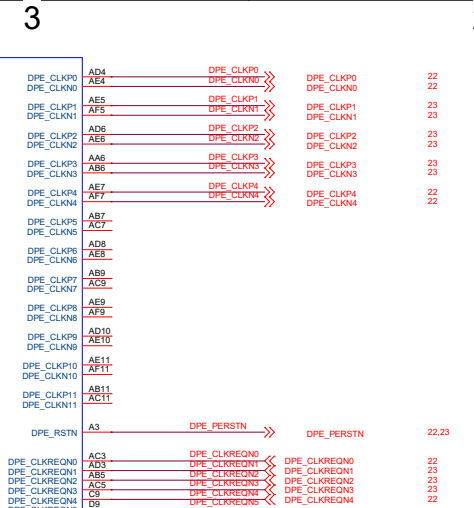
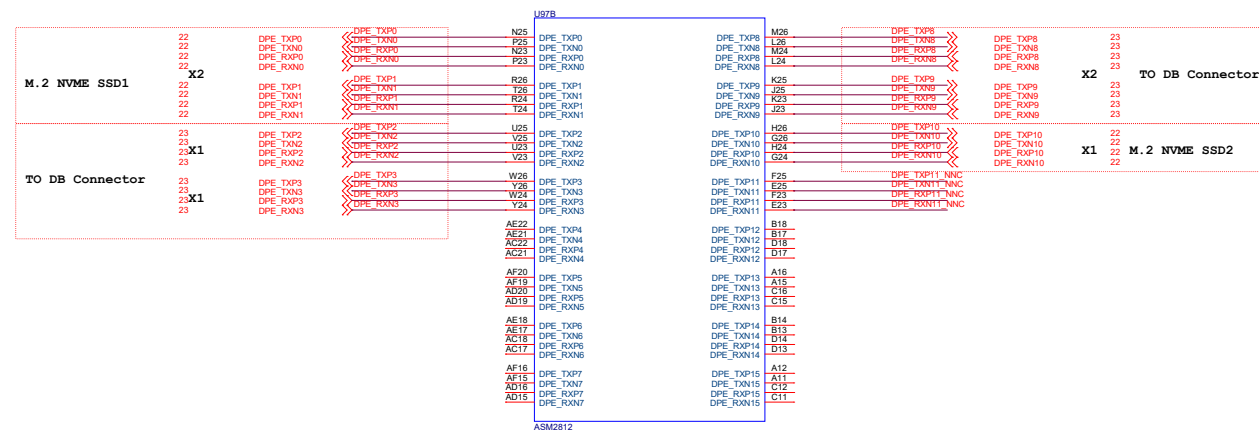
B

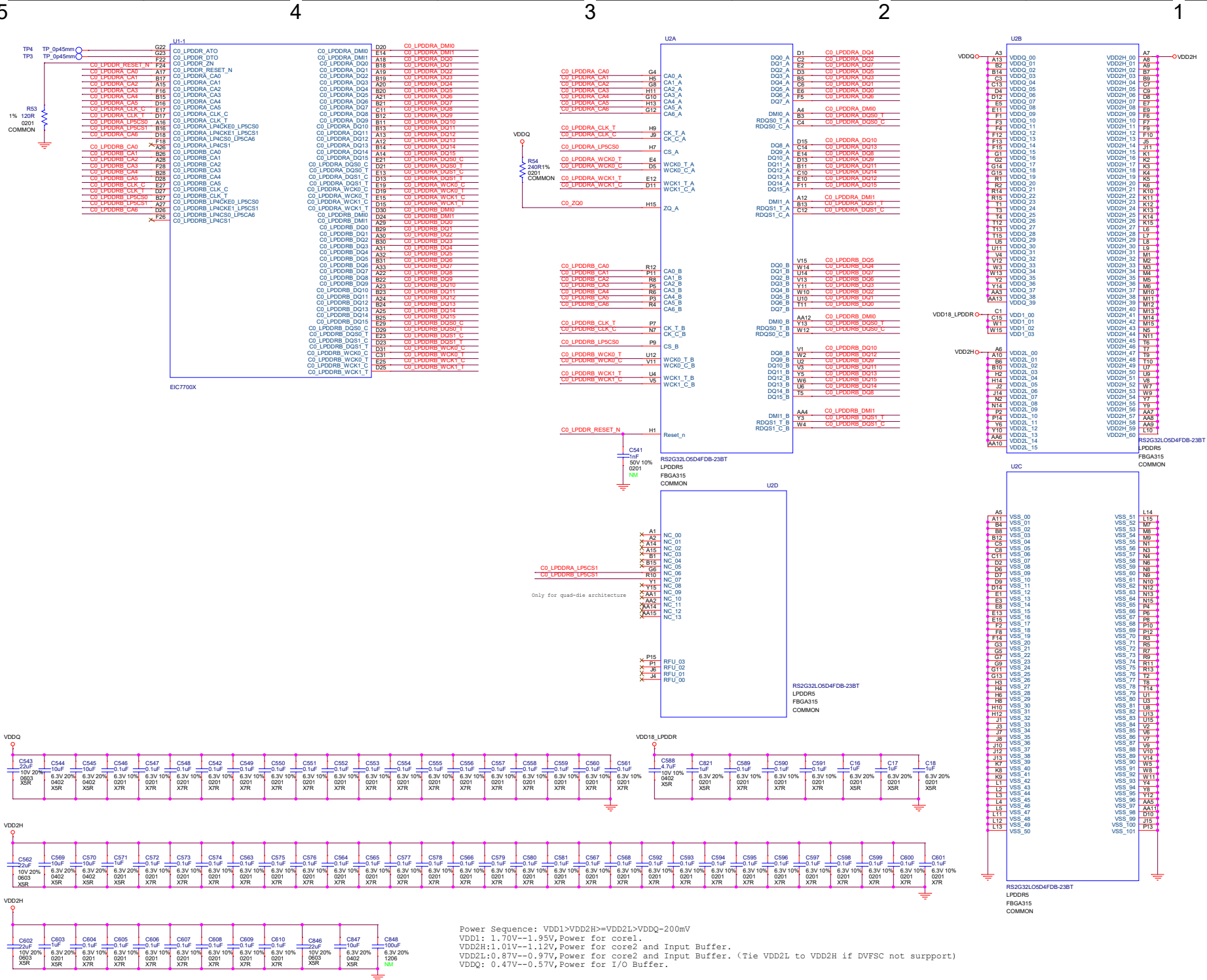
A

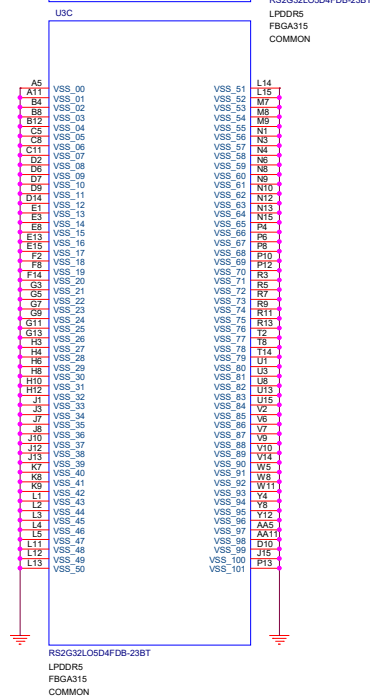
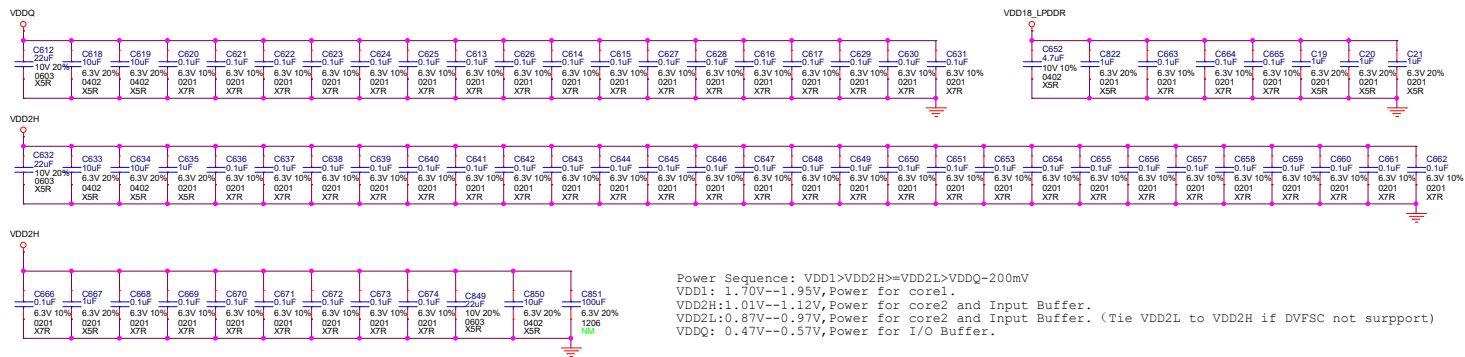
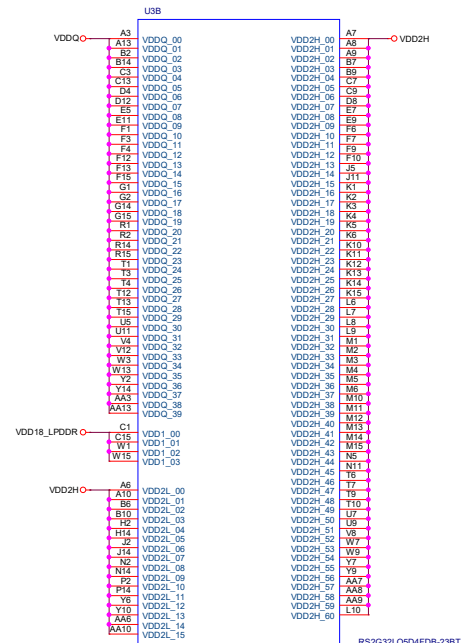
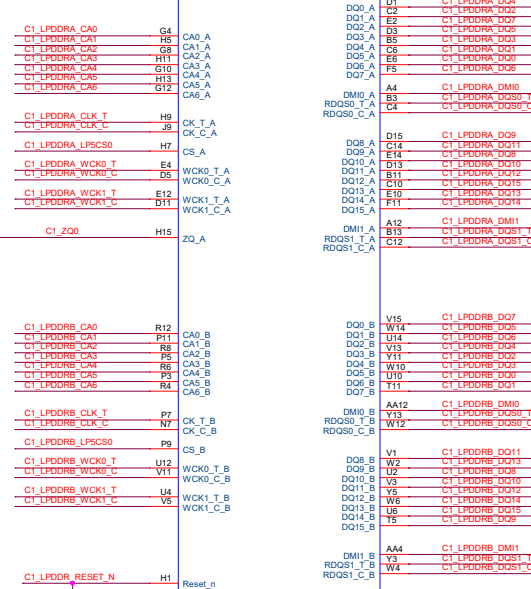


ASM2812 First 8 lane group PCIe lane configuration:

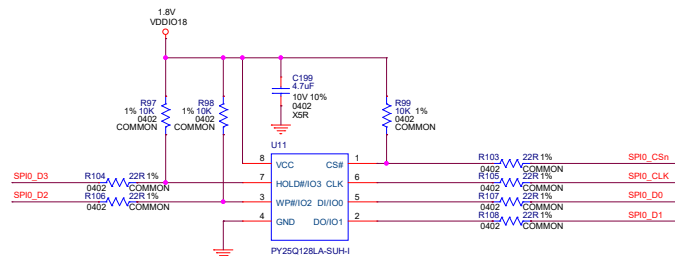
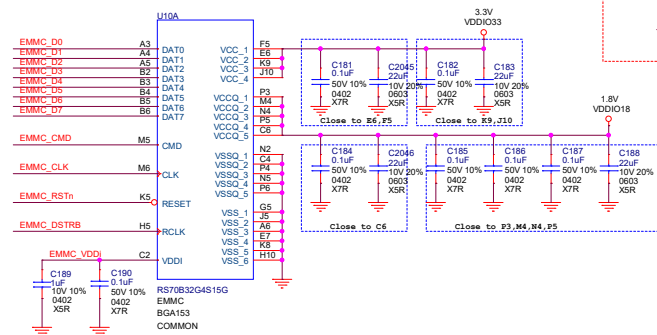
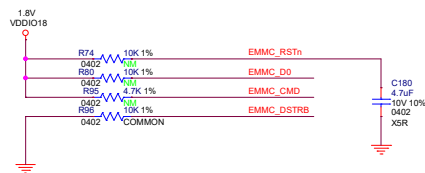
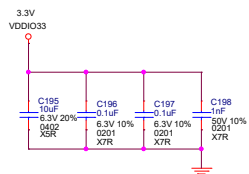
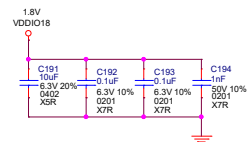
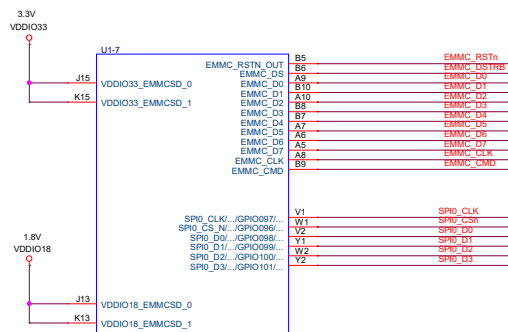
Strapping bits	LANE0	LANE1	LANE2	LANE3	LANE8	LANE9	LANE10	LANE11
GPIO5, DEBUG27, SPI_SDIO								
PECLK Mapping	LANE0	LANE1	LANE2	LANE3	LANE4	LANE5		
0 0 0	x2	x1	x1	x2	x1	x1		
0 0 1	x2	x1	x1		x4			
0 1 0		x4		x2	x1	x1		
0 1 1		x4			x4			
1 0 0				x8				



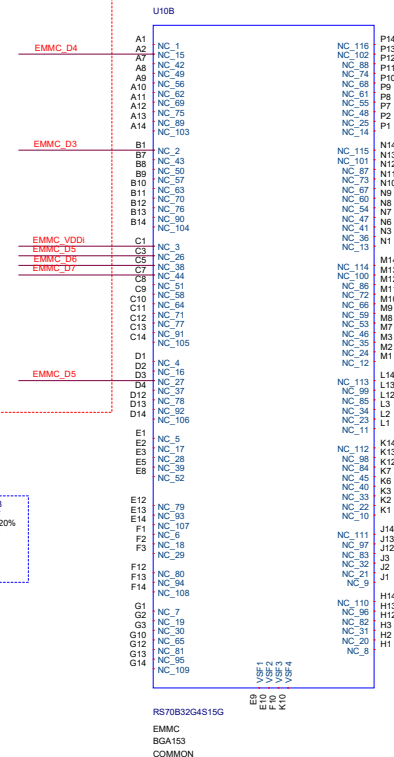




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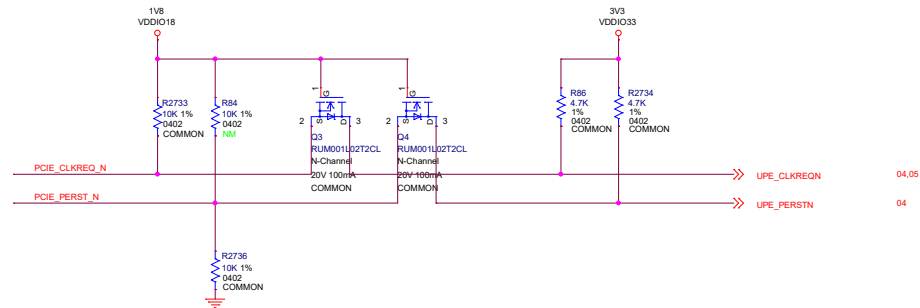
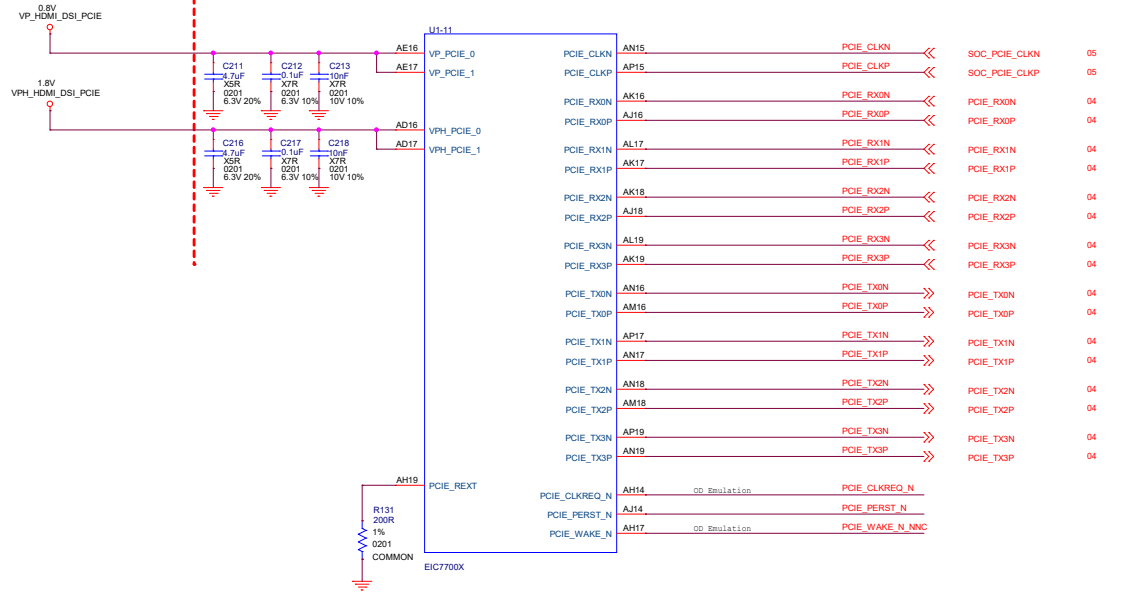


For Layout only



Close to Package

Under Package

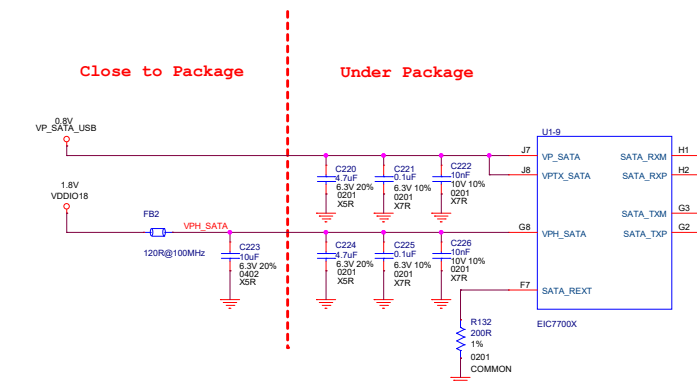
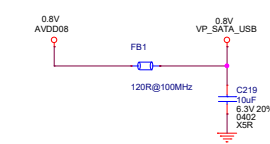
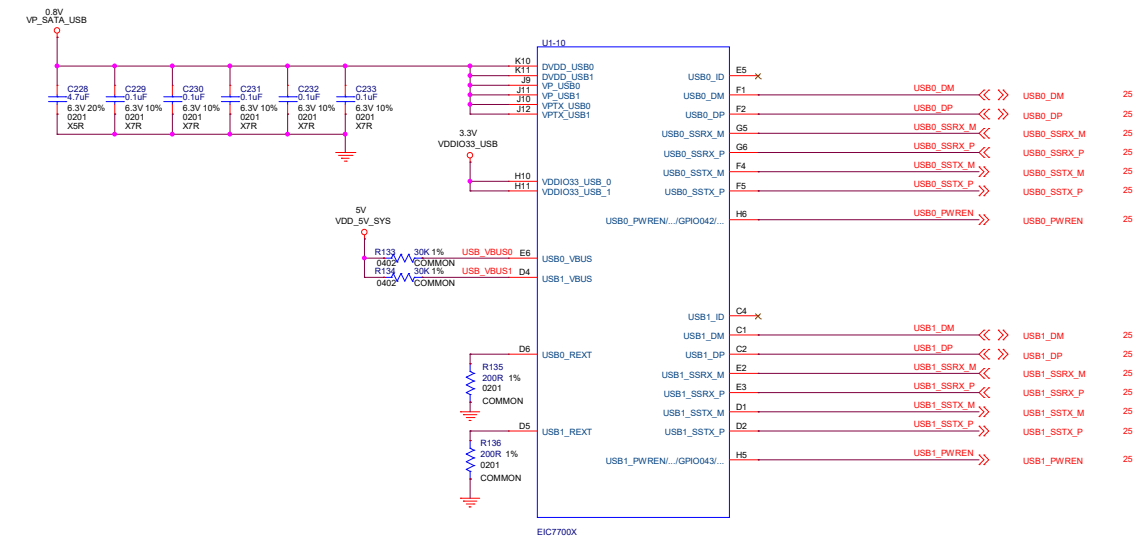
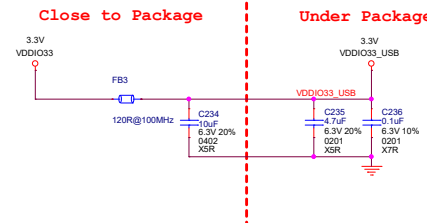


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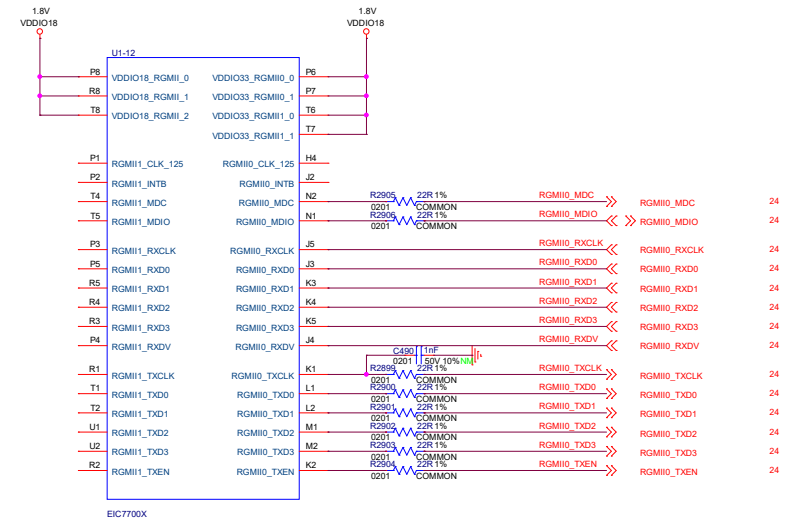
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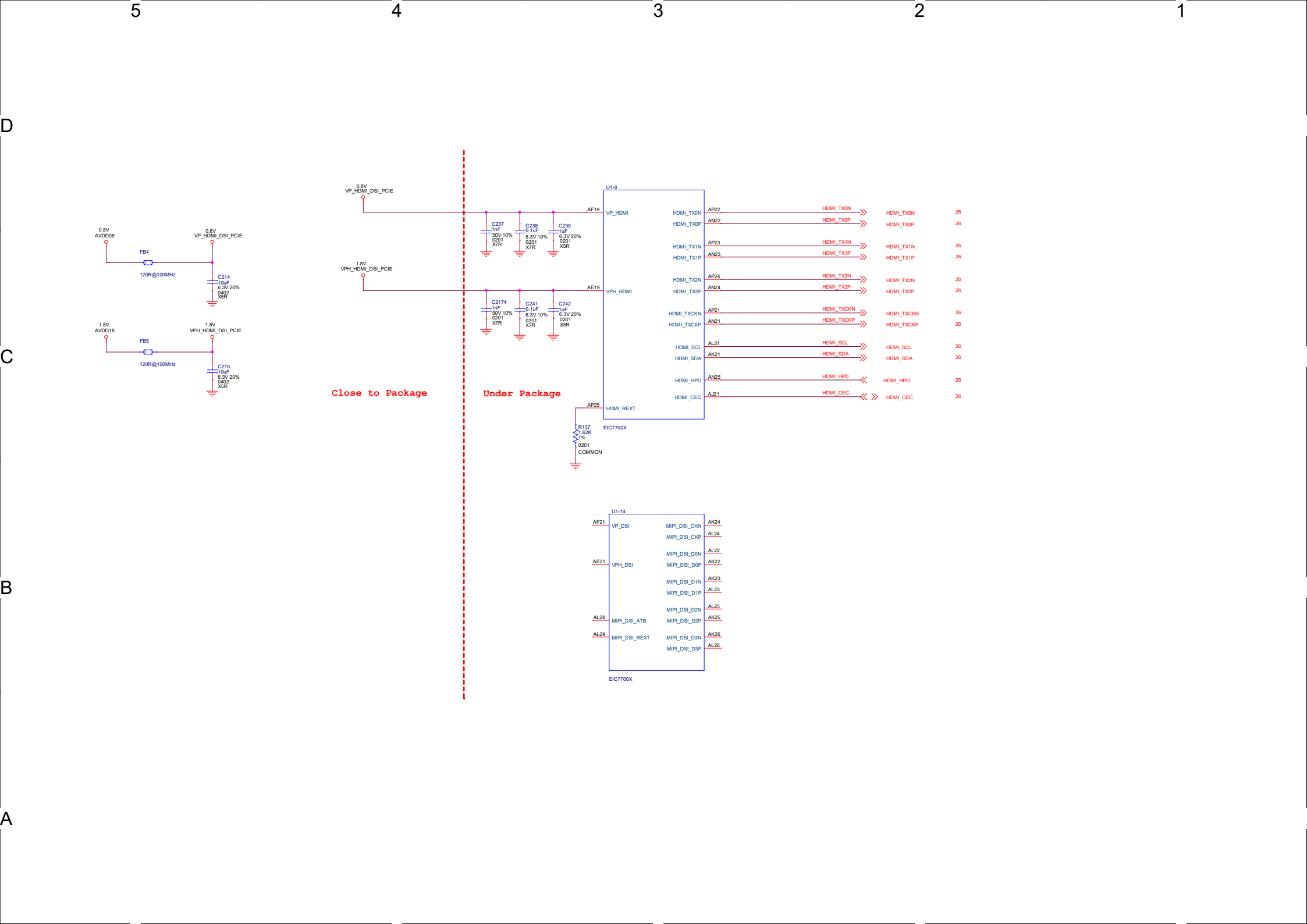
B

A

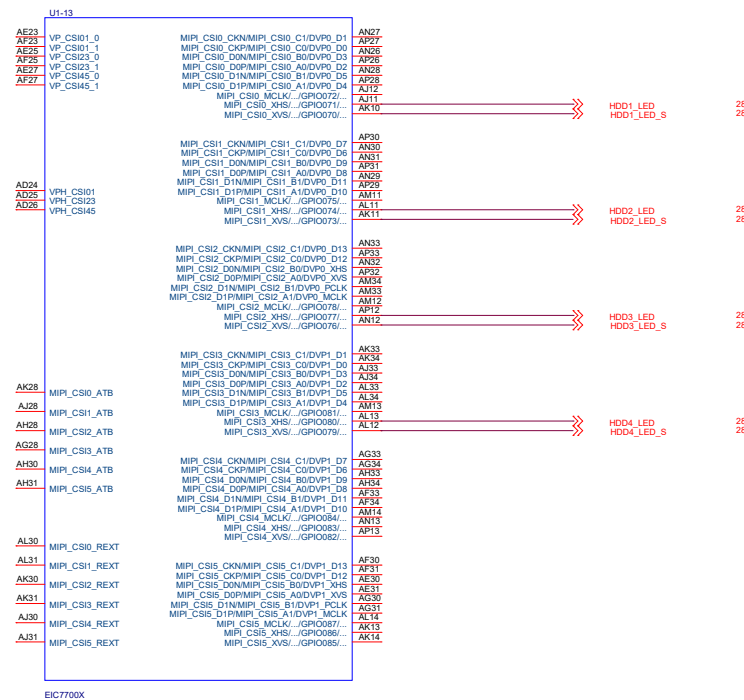


A





A

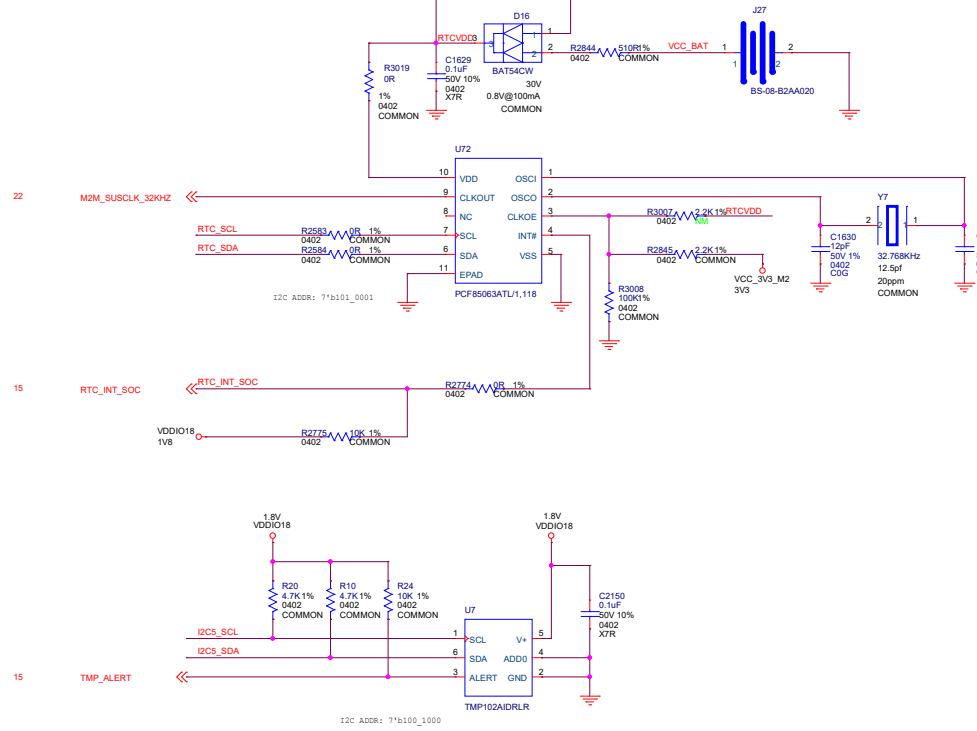
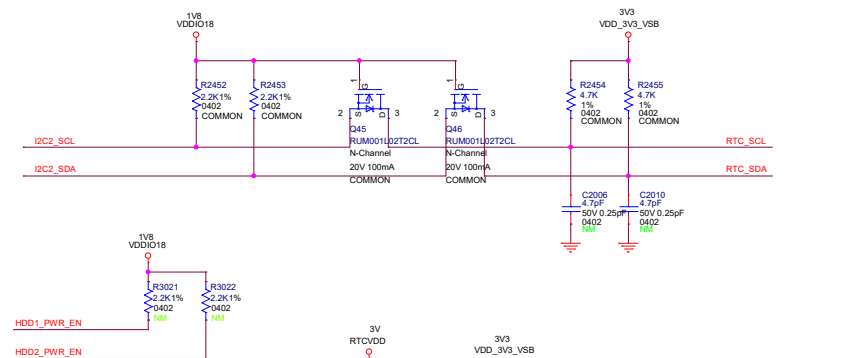
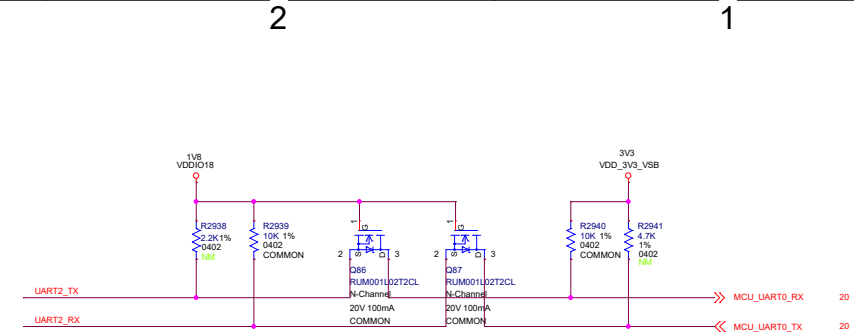
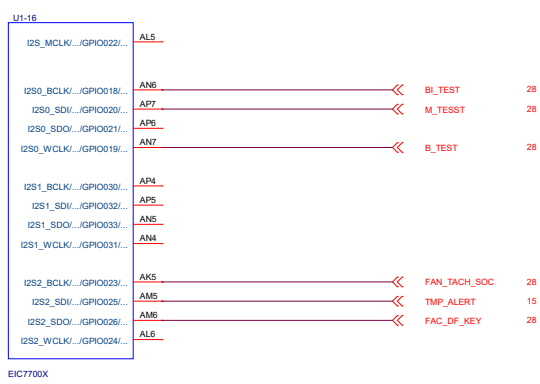
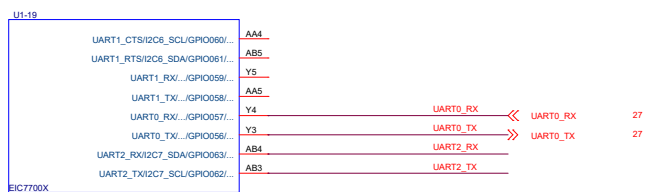
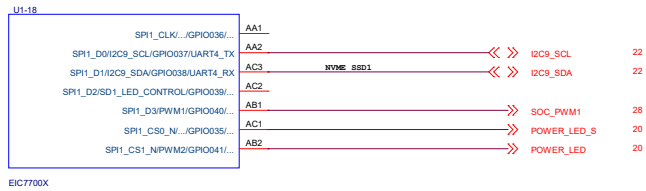
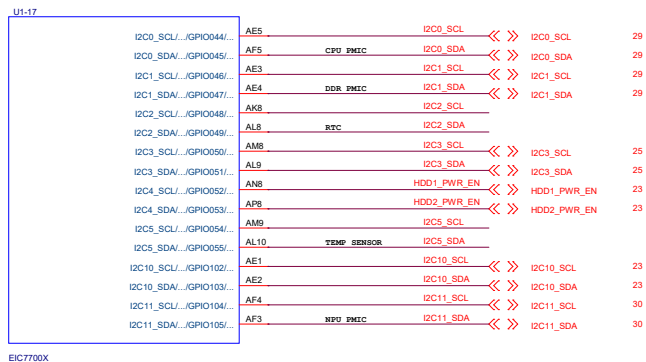
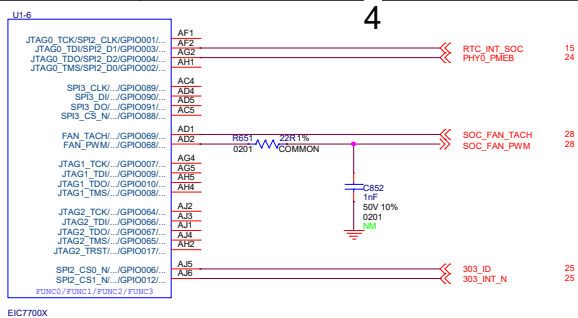


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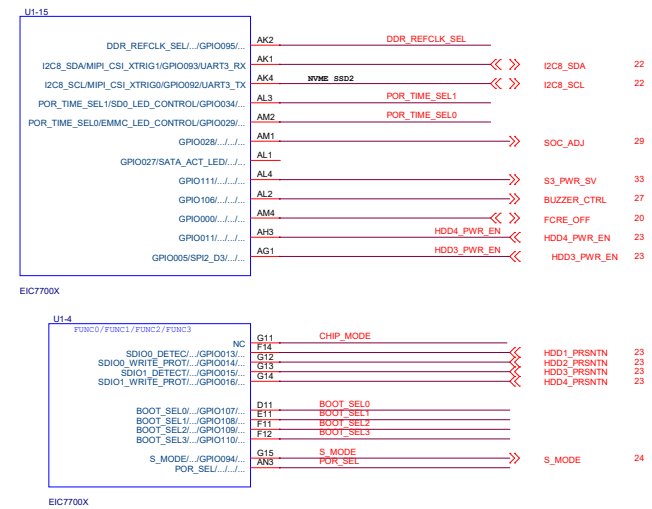
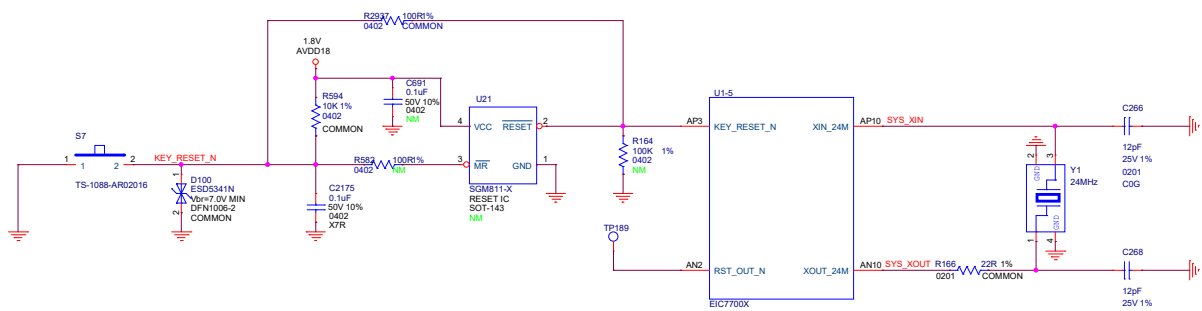
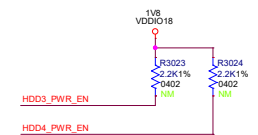
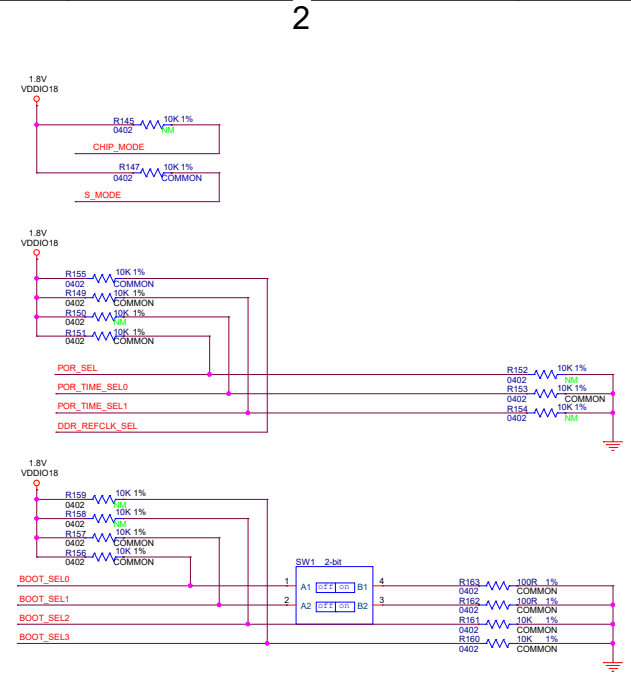
C

B

A



Strap Bit	Description	Pin	Default	Comments
CHIP MODE MODE SET3 MODE SET2 MODE SET1 MODE SET0	0xxxx : function mode 10000 : test0 10001 : test1 10010 : test2 10011 : test3 . . 11111 : test15	G11: CHIP_MODE G14: MODE_SET3 G13: MODE_SET2 G12: MODE_SET1 F14: MODE_SET0	0xxxx	
POR SELECT	0 : Bypass Internal POR 1 : Internal POR	AN3: POR_SEL	1	
POR WAIT TIME[1] POR WAIT TIME[0]	00 : 2560us 01 : 20480us 10 : 81920us 11 : 327680us	A13: GPIO34 AM2: GPIO29	10	
BOOT SELECT 3 BOOT SELECT 2 BOOT SELECT 1 BOOT SELECT 0	if OTP security bit = 1 bits : boot cpu : first boot : second boot xx00 : SCPU : RCM : UART xx01 : SCPU : RCM : eMMC xx10 : SCPU : RCM : SPI NOR xx11 : SCPU : RCM : USB if OTP security bit = 0 bits : boot cpu : first boot : second boot 0000 : SCPU : RCM : UART 0001 : SCPU : RCM : eMMC 0010 : SCPU : RCM : SPI NOR 0011 : SCPU : RCM : USB 1x00 : U84 : SPI NOR : UART 1x01 : U84 : SPI NOR : eMMC 1x10 : U84 : SPI NOR : SPI NOR 1x11 : U84 : SPI NOR : USB	F12: BOOT SELECT 3 F11: BOOT SELECT 2 E11: BOOT SELECT 1 D11: BOOT SELECT 0	0010	
S_MODE	1 : Single Die mode 0 : Dual Die mode	G15: S_MODE	1	This design is only for single die mode
LPDDR_REF_CLK	0 : Backup clock 1 : Main clock	AN2:GPIO95	1	This design is main clock only



D

C

B

A

